

MOTIVIC THETA LIFTING AND BEILINSON'S CONJECTURES

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ABSTRACT. Let V be a quadratic space over $\mathbb{Q}$ of real signature $(n, 2)$, and let $\mathrm{Sh}_{\mathrm{SO}(V)}$ be the associated orthogonal Shimura variety. We describe a genus-two cohomological construction of a Chow-valued theta function and a higher-Chow-valued class

$$\begin{aligned}\Theta^{\mathrm{mot},0} &\in H^0(X_2^{\mathrm{tor}}, \omega_2^{1+n/2}) \otimes \mathrm{CH}^2(\mathrm{Sh}_{\mathrm{SO}(V)})_{\mathbb{Q}}, \\ \Theta^{\mathrm{mot},1} &\in H^1(X_2^{\mathrm{tor}}, \omega_2^{1+n/2}) \otimes \mathrm{CH}^2(\mathrm{Sh}_{\mathrm{SO}(V)}, 1)_{\mathbb{Q}}.\end{aligned}$$

Here X_2^{tor} is a toroidal compactification of the Siegel threefold, and ω_2 denotes the canonical extension of the Hodge line bundle. The construction is organized by a map from a boundary local-cohomology complex on the Siegel threefold to a modified Gersten complex on $\mathrm{Sh}_{\mathrm{SO}(V)}$. Serre duality realizes the induced cohomological map as the motivic class $\Theta^{\mathrm{mot},1}$. Its Beilinson regulator is identified, through a motivic Thom property, with the $(1, 3)$ -component of a joint Kudla–Millson theta class. For $n = 2$, this gives a motivic theta lift of coherent cohomology classes for the pair $(\mathrm{Sp}_4, \mathrm{SO}(V))$. Pairing its regulator with the corresponding geometric theta lift yields a doubling zeta integral, which unfolds to a near-central standard L -value, up to local and normalization factors.

A CONVENTION FOR THESE NOTES

Centers? Not today!

ORTHOGONAL SIDE

GSpin	Spin	GO
GSO	O	SO



SYMPLECTIC SIDE

Sp	GSp
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Central kernels, similitude factors, and connected components are suppressed.

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This is ongoing joint work with Wenxuan Qi, Liang Xiao, and Yichao Zhang.

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1 Introduction

1.1 The guiding problem

Classical theta lifting starts with a reductive dual pair and produces automorphic forms. Arithmetic theta lifting replaces scalar Fourier coefficients by algebraic cycles or arithmetic classes. Here we construct cohomological theta classes with coefficients in higher Chow groups and compare their Beilinson regulators with automorphic L -values.

Let V be a quadratic space over $\mathbb{Q}$ of real signature $(n, 2)$. We work with the pair

$$(\mathrm{Sp}_4, \mathrm{SO}(V)).$$

The construction proceeds by comparing the boundary local-cohomology complex of the Siegel threefold with a modified Gersten complex on the orthogonal Shimura variety. Passing to cohomology and applying Serre duality yields both the Chow-valued theta series $\Theta^{\mathrm{mot},0}$ and its higher-Chow-valued companion $\Theta^{\mathrm{mot},1}$. In particular, the higher Chow class arises from this comparison, rather than from a choice of modular units.

Main results

- (1) A genus-two Chow-valued theta function

$$\Theta^{\mathrm{mot},0}(\tau) = \sum_{T \geq 0} Z(T) q^T.$$

- (2) A canonical higher-Chow-valued cohomology class $\Theta^{\mathrm{mot},1}$, characterized by Serre duality.

- (3) A Hodge–Deligne inner product formula: we express the pairing between the Beilinson regulator of a motivic theta lift and a geometric theta lift as a doubling zeta integral, and evaluate it in terms of a near-central standard L -value.

Remark 1.1 (Height pairings and regulator pairings)

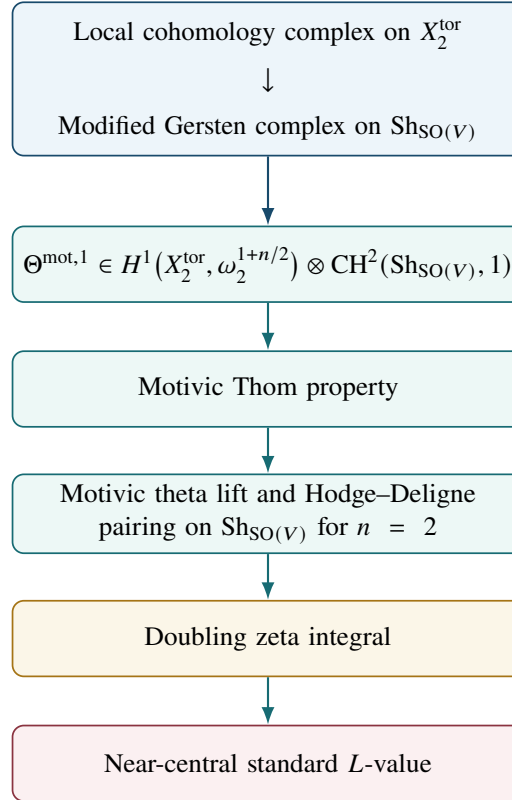
The classical Rallis formula pairs automorphic forms, while arithmetic theta lifting pairs Chow classes by Beilinson–Bloch heights. Here we instead work with higher Chow classes, in the motivic cohomology group

$$H_{\mathrm{mot}}^{2m-1}(X, \mathbb{Q}(m))_{\mathbb{Q}} \simeq \mathrm{CH}^m(X, 1)_{\mathbb{Q}}.$$

Thus the scalar pairing is obtained after applying the Beilinson regulator, rather than from a Beilinson–Bloch height pairing. This is why the final identity is called a *Hodge–Deligne inner product formula*; compare [Bei85b, Blo86, BG02].

1.2 Outline of the construction

The construction may be summarized as follows.



1.3 Main statements

Theorem 1.2 (The two genus-two motivic theta classes; see Theorem 4.16)

The construction gives

$$\Theta^{\text{mot},0} \in H^0(X_2^{\text{tor}}, \omega_2^{1+n/2}) \otimes \text{CH}^2(\text{Sh}_{\text{SO}(V)})_{\mathbb{C}}, \quad \Theta^{\text{mot},0} = \sum_{T \geq 0} Z(T) q^T,$$

and

$$\Theta^{\text{mot},1} \in H^1(X_2^{\text{tor}}, \omega_2^{1+n/2}) \otimes \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1)_{\mathbb{C}}.$$

The second class is the Serre-dual tensor representing the map

$$H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \longrightarrow \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1)_{\mathbb{C}}.$$

Corollary 1.3 (Kudla's modularity conjecture)

The arithmetic theta series $\sum_{T \geq 0} Z(T) q^T$ is a Siegel modular form of weight $1 + n/2$. This gives a new proof of Kudla's modularity conjecture in genus two.

Theorem 1.4 (Motivic Thom property; see Theorem 6.22)

The Beilinson regulator of $\Theta^{\text{mot},1}$ is the $(1, 3)$ -Künneth component of the joint Kudla–Millson theta class:

$$r_D(\Theta^{\text{mot},1}) = \Theta^{\text{DB},1} = K^{(13)}.$$

Here $K^{(13)}$ is defined in Section 6.6.

Theorem 1.5 (Beilinson-type regulator formula; see Theorem 8.12)

Specialize to $n = 2$. Let π be a suitably chosen automorphic representation of $\mathrm{Sp}_4(\mathbb{A})$, put $\pi' = \pi^\vee$, and let $\Pi = \theta(\pi)$ be its global theta lift to $\mathrm{SO}(V)(\mathbb{A})$.

For suitable vectors $g \in \pi$ and $g' \in \pi'$, let

$$\omega_g \in H^2(X_2^{\mathrm{tor}}, \omega_2(-D)), \quad \omega_{g'} \in H^1(X_2^{\mathrm{tor}}, \omega_2^2(-D))$$

be the canonical coherent cohomology classes defined in Section 7.2. Let

$$\Theta^B(\omega_{g'}) \in H_B^2(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{C})$$

be the corresponding geometric theta lift constructed in Section 7.6. Then

$$\langle r_D(\Theta^{\mathrm{mot}}(\omega_g)), \Theta^B(\omega_{g'}) \rangle = (*) L(1, \Pi, \mathrm{std}),$$

where the factor $(*)$ is specified in Theorem 8.12.

Theorem 1.6 (Nonvanishing; see Theorem 8.14)

For suitable choices of the finite Schwartz function φ_f and the finite component g_f of g , the motivic theta lift $\Theta^{\mathrm{mot}}(\omega_g)$ is nonzero.

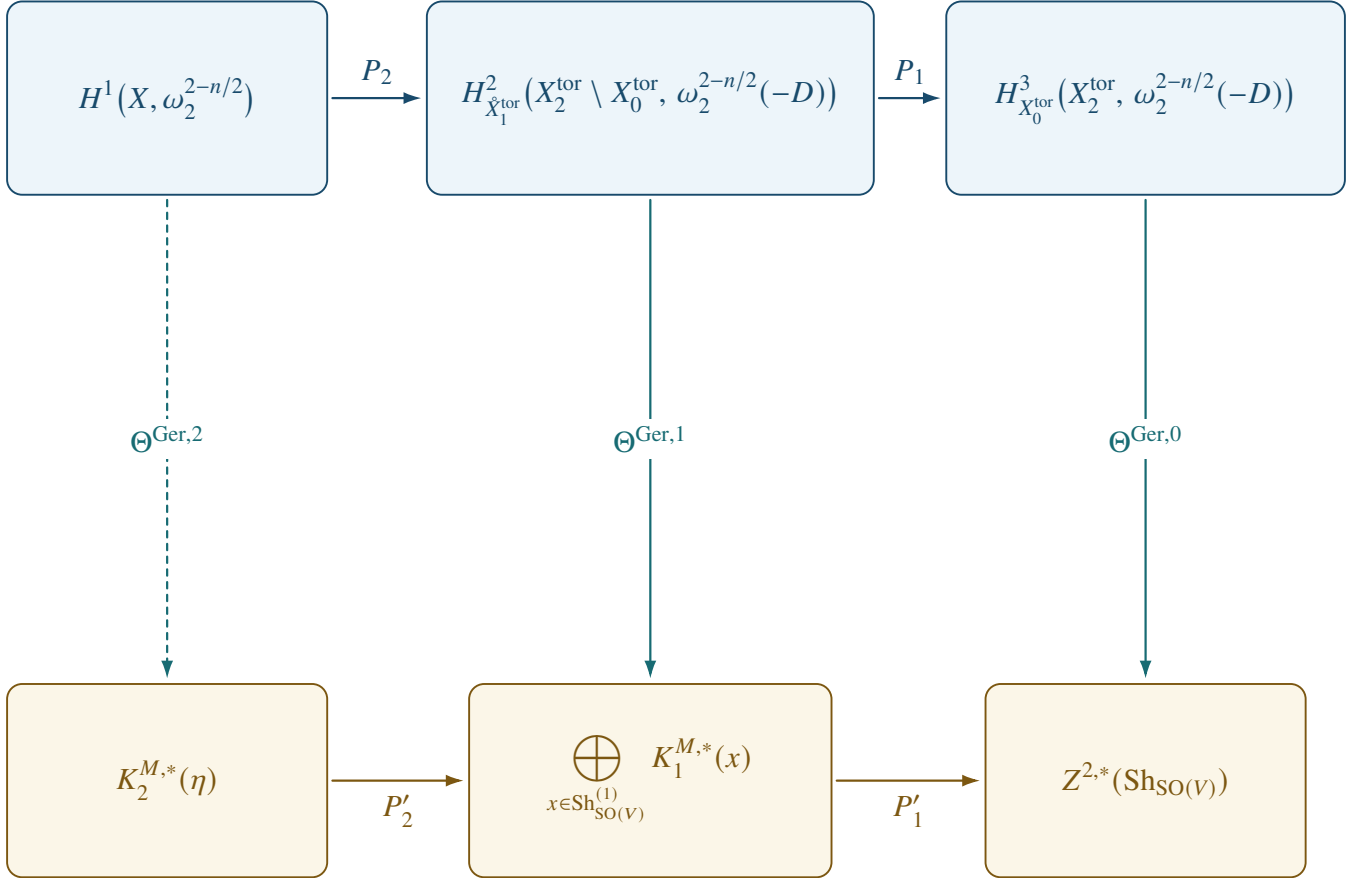
Notational convention

We often suppress the dependence of theta kernels, classes, and lifts on the finite Schwartz data φ_f when it is clear from the context.

Part 1

Construction of the motivic theta class

$$\Theta^{\text{mot},1}$$



2 Special cycles on orthogonal Shimura varieties

2.1 The orthogonal Shimura variety

Let (V, Q) have signature $(n, 2)$ and let $\Lambda \subset V$ be an integral lattice. Write

$$\mathbb{D}_V := \{W \subset V_{\mathbb{R}} : \dim W = 2, Q|_W < 0\}$$

for the space of oriented negative-definite two-planes. Suppressing connected-component and level notation, we write

$$\mathrm{Sh}_{\mathrm{SO}(V)}(\mathbb{C}) = \mathrm{SO}(\Lambda) \backslash \mathbb{D}_V^+. \quad (2.1)$$

All constructions below admit the usual adelic and finite-level refinements.

For $v \in V$ with $Q(v) > 0$, the orthogonal complement has signature $(n-1, 2)$ and gives a special divisor

$$Z(v) = \mathrm{Sh}_{\mathrm{SO}(v^\perp)} \hookrightarrow \mathrm{Sh}_{\mathrm{SO}(V)}.$$

For $\mathbf{v} = (v_1, \dots, v_r) \in \Lambda^r$, put $V_{\mathbf{v}} = \langle v_1, \dots, v_r \rangle$. The corrected intersection class is

$$Z(\mathbf{v}) := [Z(v_1) \cap \dots \cap Z(v_r)] \cdot c_1(\mathcal{L}^\vee)^{r-\dim V_{\mathbf{v}}} \in \mathrm{CH}^r(\mathrm{Sh}_{\mathrm{SO}(V)})_{\mathbb{Q}}, \quad (2.2)$$

where $\mathcal{L}$ is the tautological line bundle.

2.2 The Chow-valued theta function

For $T \in \mathrm{Sym}_r(\mathbb{Q})_{\geq 0}$ define

$$Z(T) := \sum_{\substack{\mathbf{v} \in \Lambda^r \\ Q(\mathbf{v})=T}} Z(\mathbf{v}), \quad q^T = e^{2\pi i \mathrm{Tr}(T\tau)}. \quad (2.3)$$

Kudla's generating series is

$$\Theta^q(\tau) := \sum_{T \geq 0} Z(T) q^T. \quad (2.4)$$

It is expected, and is known in the relevant formal or cohomological senses, to be a Siegel modular form of genus r and weight $1 + n/2$ with coefficients in $\mathrm{CH}^r(\mathrm{Sh}_{\mathrm{SO}(V)})_{\mathbb{Q}}$; see [KM90, Kud97, Bor98, Bor99, Zha09, BWR15, BR25].

Remark 2.1

When n is odd, the weight $1 + n/2$ is half-integral. In the general statement the modular object should be placed on the appropriate metaplectic stack (or expressed in an equivalent Weil-representation-valued form). In the genus-two calculation we write scalar powers of ω_2 to keep the geometry visible.

2.3 The genus-two target

For the main construction, the Siegel variable has genus two. Thus the arithmetic theta function takes values in codimension two:

$$\Theta^{\mathrm{mot},0} := \Theta^q \in H^0(X_2^{\mathrm{tor}}, \omega_2^{1+n/2}) \otimes \mathrm{CH}^2(\mathrm{Sh}_{\mathrm{SO}(V)})_{\mathbb{Q}}. \quad (2.5)$$

The new object is the degree-one companion

$$\Theta^{\mathrm{mot},1} \in H^1(X_2^{\mathrm{tor}}, \omega_2^{1+n/2}) \otimes \mathrm{CH}^2(\mathrm{Sh}_{\mathrm{SO}(V)}, 1)_{\mathbb{Q}}.$$

The genus-two construction is forced by the boundary geometry of X_2 . Before turning to it, we record the genus-one construction.

3 The genus-one construction

Let

$$Y_\Gamma = \Gamma \backslash \mathbb{H} \hookrightarrow X_\Gamma = Y_\Gamma \coprod \{\text{cusps}\}, \quad D = \{\text{cusps}\}. \quad (3.1)$$

Kudla's modularity conjecture in genus one amounts to constructing

$$\Theta(\tau) \in \text{CH}^1(\text{Sh}_{\text{SO}(V)})_{\mathbb{C}} \otimes H^0(X_\Gamma, \omega^{1+n/2}). \quad (3.2)$$

The Kodaira–Spencer isomorphism gives

$$\omega^2 \simeq \Omega_{X_\Gamma}^1(\log D) \simeq \Omega_{X_\Gamma}^1(D), \quad \omega^2(-D) \simeq \Omega_{X_\Gamma}^1. \quad (3.3)$$

Consequently, Serre duality identifies (3.2) with the construction of a map

$$\theta : H^1(X_\Gamma, \omega^{1-n/2}(-D)) \longrightarrow \text{CH}^1(\text{Sh}_{\text{SO}(V)})_{\mathbb{C}}. \quad (3.4)$$

Proposition 3.1 (The compactified cohomology as a cokernel)

There is a canonical identification

$$H^1(X_\Gamma, \omega^{1-n/2}(-D)) \simeq \text{Coker}\left(H^0(Y_\Gamma, \omega^{1-n/2}) \longrightarrow H_{\{\text{cusps}\}}^1(X_\Gamma, \omega^{1-n/2}(-D))\right). \quad (3.5)$$

Proof. The spectral sequence for the stratification $X_\Gamma = Y_\Gamma \coprod \{\text{cusps}\}$ has the following E_1 -page:

$$E_1\text{-page} \quad \begin{array}{ccc} H^1(Y_\Gamma, \omega^{1-n/2}(-D)) & H_{\{\text{cusps}\}}^2(X_\Gamma, \omega^{1-n/2}(-D)) & \\ H^0(Y_\Gamma, \omega^{1-n/2}(-D)) & H_{\{\text{cusps}\}}^1(X_\Gamma, \omega^{1-n/2}(-D)) & \end{array} \implies H^1(X_\Gamma, \omega^{1-n/2}(-D)) \quad (3.6)$$

Since Y_Γ is affine,

$$H^1(Y_\Gamma, \omega^{1-n/2}(-D)) = 0, \quad H^0(Y_\Gamma, \omega^{1-n/2}(-D)) = H^0(Y_\Gamma, \omega^{1-n/2}),$$

and

$$H_{\{\text{cusps}\}}^2(X_\Gamma, \omega^{1-n/2}(-D)) = 0.$$

Thus the degree-one part of (3.6) is

$$H^0(Y_\Gamma, \omega^{1-n/2}) \longrightarrow H_{\{\text{cusps}\}}^1(X_\Gamma, \omega^{1-n/2}(-D)) \longrightarrow H^1(X_\Gamma, \omega^{1-n/2}(-D)) \longrightarrow 0,$$

which gives (3.5). $\square$

Remark 3.2

The first term in (3.5) is the space of weakly holomorphic modular forms of weight $1 - n/2$. The map takes a weakly holomorphic modular form to its polar part at the cusps.

On the orthogonal side,

$$\text{CH}^1(\text{Sh}_{\text{SO}(V)})_{\mathbb{C}} = \text{Coker}\left(\text{Mero}(\text{Sh}_{\text{SO}(V)}) \xrightarrow{\text{div}} Z^1(\text{Sh}_{\text{SO}(V)})\right) \quad (3.7)$$

$$= \text{Coker}\left(\text{Mero}^*(\text{Sh}_{\text{SO}(V)}) \xrightarrow{(\text{div}, \text{id})} Z^{1,*}(\text{Sh}_{\text{SO}(V)})\right), \quad (3.8)$$

where

$$\text{Mero}^*(\text{Sh}_{\text{SO}(V)}) = \text{Mero}(\text{Sh}_{\text{SO}(V)}) \oplus \mathbb{C}[\mathcal{L}^\vee], \quad Z^{1,*}(\text{Sh}_{\text{SO}(V)}) = Z^1(\text{Sh}_{\text{SO}(V)}) \oplus \mathbb{C}[\mathcal{L}^\vee]. \quad (3.9)$$

The additional summand records the weight.

The divisor formula for the Borchers product gives the comparison

$$\begin{array}{ccc}
 H^0(Y_\Gamma, \omega^{1-n/2}) & \xrightarrow{\text{Polar part}} & H^1_{\{\text{cusps}\}}(X_\Gamma, \omega^{1-n/2}(-D)) \\
 \downarrow f \mapsto \Psi_f & & \downarrow q^{-m} \quad \downarrow 1 \\
 \text{Mero}^*(\text{Sh}_{\text{SO}(V)}) & \xrightarrow{(\text{div}, \text{id})} & Z^{1,*}(\text{Sh}_{\text{SO}(V)}) \\
 & & \downarrow [Z(m)] \quad \downarrow \mathcal{L}^\vee
 \end{array} \tag{3.10}$$

Remark 3.3

Passing to cokernels in (3.10) gives the map (3.4). Under Serre duality, this recovers $\Theta(\tau)$ in (3.2), with

$$\Theta(\tau) = \sum_{m \geq 0} Z(m) q^m. \tag{3.11}$$

Equivalently, the comparison gives a new cohomological interpretation of Borchers's *The Gross–Kohnen–Zagier theorem in higher dimensions* [Bor99, Theorem 4.5].

4 The genus-two cohomological construction

4.1 The Siegel threefold and its compactifications

Let

$$X = \Gamma \backslash \mathbb{H}_2$$

be the Siegel threefold. We write

$$X_2^{\min} := X^{\min}, \quad X_2^{\text{tor}} := X^{\text{tor}}$$

for its Baily–Borel compactification and a toroidal compactification. The stratifications used in the construction are

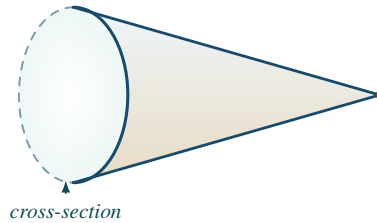
$$\begin{array}{ccccc}
 X_2^{\text{tor}} & \supset & X_1^{\text{tor}} & \supset & X_0^{\text{tor}} \\
 \downarrow \pi & & \downarrow \pi & & \downarrow \pi \\
 X_2^{\min} & \supset & X_1^{\min} & \supset & X_0^{\min}.
 \end{array} \tag{4.1}$$

Here X_1^{tor} is the universal elliptic curve over the modular curve, $X_1^{\min}$ is the modular curve, and $X_0^{\min}$ is a finite set. Put

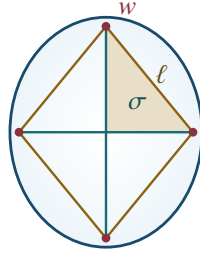
$$D = X_2^{\text{tor}} \setminus X, \quad \mathring{X}_i^{\text{tor}} = X_i^{\text{tor}} \setminus X_{i-1}^{\text{tor}}, \quad \mathring{X}_i^{\min} = X_i^{\min} \setminus X_{i-1}^{\min}. \tag{4.2}$$

For each $x \in X_0^{\min}$, the fiber $\pi^{-1}(x)$ in X_0^{tor} is the boundary of the toric variety associated with a suitable cone decomposition of $\text{Sym}_{2 \times 2}(\mathbb{R})^{\geq 0}$. The chosen decomposition is illustrated below.

cone decomposition



decomposition of the cross-section



4.2 The Cousin spectral sequence and local cohomology

Proposition 4.1 (The local-cohomology complex)

The Cousin spectral sequence gives the complex

$$H^1(X, \omega_2^{2-n/2}) \xrightarrow{P_2} H_{X_1^{\text{tor}}}^2(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) \xrightarrow{P_1} H_{X_0^{\text{tor}}}^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)). \quad (4.3)$$

Its cohomology gives

$$H^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq \text{Coker } P_1, \quad (4.4)$$

and

$$H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq \frac{\ker P_1}{\text{Im } P_2}. \quad (4.5)$$

Proof. The filtration

$$X_2^{\text{tor}} \supset X_1^{\text{tor}} \supset X_0^{\text{tor}}$$

gives the following Cousin spectral sequence:

$$\begin{array}{|c|c|c|c|c|} \hline & H^3(X, \omega_2^{2-n/2}) = 0 & & & \\ \hline E_1\text{-page} & H^2(X, \omega_2^{2-n/2}) = 0 & H_{X_1^{\text{tor}}}^3(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) = 0 & & \\ \hline & H^1(X, \omega_2^{2-n/2}) & H_{X_1^{\text{tor}}}^2(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) & H_{X_0^{\text{tor}}}^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) & \implies H^\bullet(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \\ \hline & H^0(X, \omega_2^{2-n/2}) & H_{X_1^{\text{tor}}}^1(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) & H_{X_0^{\text{tor}}}^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) = 0 & \\ \hline \end{array} \quad (4.6)$$

Here the vanishing on the left is

$$H^i(X, \omega_2^{2-n/2}) = 0 \quad (i \geq 2), \quad (4.7)$$

which is the genus-two case of the vanishing range proved in [Fan25, Theorem 5.10]

$$H^i(X_g, \omega_g^{g-n/2}) = 0 \quad \text{for } i > \frac{g(g+1)}{2} - 2.$$

□

Let

$$\iota_1 : \hat{X}_1^{\text{tor}} \hookrightarrow X_2^{\text{tor}} \setminus X_0^{\text{tor}}$$

be the locally closed immersion, and let $\mathcal{I}$ be its ideal sheaf.

Proposition 4.2 (The local-cohomology calculation)

(1) One has

$$H_{X_0^{\text{tor}}}^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) = 0, \quad (4.8)$$

and

$$H_{X_0^{\text{tor}}}^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq \Gamma \backslash \bigoplus_{x \in X_0^{\text{min}}} \mathbb{C}[-(\text{Sym}_{2 \times 2}(\mathbb{R})^{\geq 0})^\vee], \quad (4.9)$$

where $\text{Sym}_{2 \times 2}(\mathbb{R})^{\leq 0}$ is the space of negative-semidefinite symmetric 2×2 matrices and Γ is the group defining the level of the Siegel threefold.

(2) One has

$$H_{X_1^{\text{tor}}}^2(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq \bigoplus_{m \geq 1} H^0(\hat{X}_1^{\text{min}}, R^1 \pi_* \kappa^m \otimes \omega_1^{1-n/2}(-D)). \quad (4.10)$$

Proof. (1) Let $\mathcal{F} = \omega^{2-n/2}(-D)$. We first compute the local cohomology above a single cusp, suppressing the action of the level group. The Čech spectral sequence for cohomology with supports has the following E_1 -page:

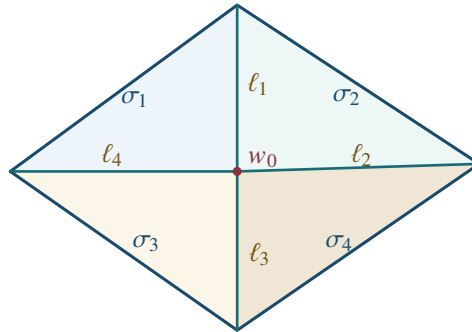
 E_1 -page

3	$\bigoplus_i H_{\mathcal{O}_{\ell_i}}^3(X_2^{\text{tor}}, \mathcal{F})$	$\bigoplus_i H_{\mathcal{O}_{w_i}}^3(X_2^{\text{tor}}, \mathcal{F})$	
2	$\bigoplus_i H_{\mathcal{O}_{\ell_i}}^2(X_2^{\text{tor}}, \mathcal{F})$	$\bigoplus_i H_{\mathcal{O}_{w_i}}^2(X_2^{\text{tor}}, \mathcal{F})$	$\Rightarrow H_{X_0^{\text{tor}}}^i(X_2^{\text{tor}}, \mathcal{F})$
1	$\bigoplus_i H_{\mathcal{O}_{\ell_i}}^1(X_2^{\text{tor}}, \mathcal{F})$	$\bigoplus_i H_{\mathcal{O}_{w_i}}^1(X_2^{\text{tor}}, \mathcal{F})$	
	-1	0	

(4.11)

It therefore suffices to compute the local cohomology groups $H_{\mathcal{O}_{\ell_i}}^\bullet(X_2^{\text{tor}}, \mathcal{F})$ and $H_{\mathcal{O}_{w_i}}^\bullet(X_2^{\text{tor}}, \mathcal{F})$.

We begin with $H_{\mathcal{O}_{w_0}}^\bullet(X_2^{\text{tor}}, \mathcal{F})$. Near w_0 , the cone decomposition has the following form:



Thus

$$\overline{\mathcal{O}}_{w_0} = \mathcal{O}_{w_0} \cup \bigcup_{i=1}^4 \mathcal{O}_{\ell_i} \cup \bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}. \quad (4.12)$$

The local cohomology group $H_{\mathcal{O}_{w_0}}^\bullet(X_2^{\text{tor}}, \mathcal{F})$ can be computed using the following Cousin spectral sequence:

E_1 -page

$$\begin{array}{ccc}
 H_{\mathcal{O}_{w_0}}^1(X_2^{\text{tor}}, \mathcal{F}) & H_{\bigcup_{i=1}^4 \mathcal{O}_{\ell_i}}^2\left(X_2^{\text{tor}} \setminus \bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}, \mathcal{F}\right) & H_{\bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}}^3(X_2^{\text{tor}}, \mathcal{F}) \\
 & H_{\bigcup_{i=1}^4 \mathcal{O}_{\ell_i}}^1\left(X_2^{\text{tor}} \setminus \bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}, \mathcal{F}\right) & H_{\bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}}^2(X_2^{\text{tor}}, \mathcal{F}) \implies H_{\mathcal{O}_{w_0}}^i(X_2^{\text{tor}}, \mathcal{F}) \\
 & & H_{\bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}}^1(X_2^{\text{tor}}, \mathcal{F})
 \end{array}
 \tag{4.13}$$

The individual terms are as follows:

(a)

$$H_{\mathcal{O}_{w_0}}^1(X_2^{\text{tor}}, \mathcal{F}) \cong H_{\mathcal{O}_{w_0}}^1(X_\Sigma, \mathcal{O}) \cong \mathbb{C}[-w_0^\vee], \tag{4.14}$$

(b)

$$H_{\bigcup_{i=1}^4 \mathcal{O}_{\ell_i}}^i(X_2^{\text{tor}} \setminus \bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}, \mathcal{F}) \cong H_{\bigcup_{i=1}^4 \mathcal{O}_{\ell_i}}^i(X_\Sigma, \mathcal{O}) \tag{4.15}$$

$$\cong \begin{cases} 0 & i \neq 2 \\ \mathbb{C}[\bigcup_{i=1}^4 \ell_i^\vee] & i = 2 \end{cases}, \tag{4.16}$$

(c)

$$H_{\bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}}^i(X_2^{\text{tor}}, \mathcal{F}) \cong H_{\bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}}^i(X_\Sigma, \mathcal{O}) \cong \begin{cases} 0 & i \neq 3 \\ \mathbb{C}[\bigcup_{i=1}^4 \sigma_i^\vee] & i = 3 \end{cases}. \tag{4.17}$$

Thus

$$H_{\mathcal{O}_{w_0}}^3(X_2^{\text{tor}}, \mathcal{F}) \cong \text{coker}(H_{\bigcup_{i=1}^4 \mathcal{O}_{\ell_i}}^2(X_2^{\text{tor}} \setminus \bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}, \mathcal{F}) \rightarrow H_{\bigcup_{i=1}^4 \mathcal{O}_{\sigma_i}}^3(X_2^{\text{tor}}, \mathcal{F})) \tag{4.18}$$

$$\cong \mathbb{C}[-\bigcap_{i=1}^4 \sigma_i^\vee], \tag{4.19}$$

and

$$H_{\mathcal{O}_{w_0}}^2(X_2^{\text{tor}}, \mathcal{F}) = 0. \tag{4.20}$$

Similarly, we have

$$H_{\mathcal{O}_{l_1}}^3(X_2^{\text{tor}}, \mathcal{F}) \cong \mathbb{C}[-\sigma_1^\vee \cap \sigma_2^\vee], \tag{4.21}$$

$$H_{\mathcal{O}_{l_2}}^3(X_2^{\text{tor}}, \mathcal{F}) \cong \mathbb{C}[-\sigma_2^\vee \cap \sigma_4^\vee], \tag{4.22}$$

$$H_{\mathcal{O}_{l_3}}^3(X_2^{\text{tor}}, \mathcal{F}) \cong \mathbb{C}[-\sigma_3^\vee \cap \sigma_4^\vee], \tag{4.23}$$

$$H_{\mathcal{O}_{l_4}}^3(X_2^{\text{tor}}, \mathcal{F}) \cong \mathbb{C}[-\sigma_1^\vee \cap \sigma_3^\vee]. \tag{4.24}$$

$$\tag{4.25}$$

Substituting these calculations into the spectral sequence gives the desired formula.

(2) For every i , one has

$$H_{\hat{X}_1^{\text{tor}}}^i(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq H^{i-1}\left(\hat{X}_1^{\text{tor}}, \bigoplus_{m \geq 1} \frac{\mathcal{I}^{-m}}{\mathcal{I}^{-m+1}} \otimes \omega_2^{2-n/2}(-D)\right). \quad (4.26)$$

Using $\mathcal{I}^{-m}/\mathcal{I}^{-m+1} \simeq \kappa^m$ and the projection $\pi : \hat{X}_1^{\text{tor}} \rightarrow \hat{X}_1^{\text{min}}$, this becomes

$$H_{\hat{X}_1^{\text{tor}}}^i(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq H^{i-1}\left(\hat{X}_1^{\text{tor}}, \bigoplus_{m \geq 1} \kappa^m \otimes \omega_2^{2-n/2}(-D)\right). \quad (4.27)$$

Moreover,

$$H_{\hat{X}_1^{\text{tor}}}^i(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq \bigoplus_{m \geq 1} H^{i-2}\left(\hat{X}_1^{\text{min}}, R^1\pi_*\kappa^m \otimes \omega_1^{1-n/2}\right). \quad (4.28)$$

Since $\hat{X}_1^{\text{min}}$ is an affine open modular curve, only $i = 2$ remains, which proves (2). $\square$

Remark 4.3

Near the boundary $\hat{X}_1^{\text{tor}}$, an element of the nonzero local-cohomology group has the Fourier–Jacobi expansion

$$f = \sum_{m \geq 1} f_m q_{22}^{-m}, \quad (4.29)$$

with

$$f_m \in H^0\left(\hat{X}_1^{\text{min}}, R^1\pi_*\kappa^m \otimes \omega_1^{1-n/2}\right).$$

Here κ is the negative line bundle obtained by pulling back the Poincaré bundle along the diagonal of the universal elliptic curve.

4.3 The modified Gersten complex

Definition 4.4. • The modified cycle group is

$$Z^{2,*}(\text{Sh}_{\text{SO}(V)}) := Z^2(\text{Sh}_{\text{SO}(V)}) \oplus \mathbb{C}[\mathcal{L}^{\vee \otimes 2}] \oplus \mathbb{C}[\mathcal{L}^{\vee}].$$

- The modified Milnor K -group is

$$K_1^{M,*}(\eta) := \text{Mero}^*(\text{Sh}_{\text{SO}(V)}),$$

where η is the generic point of $\text{Sh}_{\text{SO}(V)}$ and Mero^* denotes the group of scalar-weight automorphic forms on $\text{Sh}_{\text{SO}(V)}$.

- The modified Milnor K -group is

$$K_2^{M,*}(\eta) = \text{Mero}^*(\text{Sh}_{\text{SO}(V)})^{\otimes 2} / \sim,$$

where $\sim$ denotes the Steinberg relation in $K_2^M(\eta)$.

Proposition 4.5

Consider the modified Gersten complex

$$K_2^{M,*}(\eta) \xrightarrow{P'_2} \bigoplus_{x \in \text{Sh}_{\text{SO}(V)}^{(1)}} K_1^{M,*}(x) \xrightarrow{P'_1} Z^{2,*}(\text{Sh}_{\text{SO}(V)}), \quad (4.30)$$

where $P'_1 := \text{div}^*$ is the modified divisor map and $P'_2 := T^*$ is the modified tame symbol. Its two cohomology groups are

$$\text{CH}^2(\text{Sh}_{\text{SO}(V)}) \simeq \text{Coker } P'_1, \quad (4.31)$$

and

$$\text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1) \simeq \frac{\ker P'_1}{\text{Im } P'_2}. \quad (4.32)$$

Proof. This follows directly from the definitions and the corresponding properties of the usual Gersten complex. $\square$

4.4 The comparison of the two complexes

We now construct a morphism between the local-cohomology complex and the modified Gersten complex:

$$\begin{array}{ccccc} H^1(X, \omega_2^{2-n/2}) & \xrightarrow{P_3} & H^2_{X_1^{\text{tor}}}(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) & \xrightarrow{P_1} & H^3_{X_0^{\text{tor}}}(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \\ \downarrow \Theta_{\text{Ger},2} & & \downarrow \Theta_{\text{Ger},1} & & \downarrow \Theta_{\text{Ger},0} \\ K_2^{M,*}(\eta) & \xrightarrow{P'_2} & \bigoplus_{x \in \text{Sh}_{\text{SO}(V)}^{(1)}} K_1^{M,*}(x) & \xrightarrow{P'_1} & Z^{2,*}(\text{Sh}_{\text{SO}(V)}). \end{array} \quad (4.33)$$

Construction 4.6 (Construction of $\Theta_{\text{Ger},0}$). Define $\Theta_{\text{Ger},0}$ by the following rank-stratified description:

$$\begin{aligned} H^3_{X_0^{\text{tor}}}(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) &\simeq \Gamma \setminus \left(\bigoplus_{x \in X_0^{\text{min}}} \mathbb{C}[-(\text{Sym}_{2 \times 2}(\mathbb{R})^{\geq 0})^\vee] \right) \longrightarrow Z^{2,*}(\text{Sh}_{\text{SO}(V)}) \\ Z^{2,*}(\text{Sh}_{\text{SO}(V)}) &\simeq Z^2(\text{Sh}_{\text{SO}(V)}) \oplus \mathbb{C}[\mathcal{L}^{\vee \otimes 2}] \oplus \mathbb{C}[\mathcal{L}^\vee] \otimes Z^1[\text{Sh}_{\text{SO}(V)}] \\ \text{rank}(T) = 2 &\longmapsto (Z(T), 0, 0), \\ \text{rank}(T) = 1 &\longmapsto (0, 0, \mathcal{L}^\vee \otimes Z(T)) \\ \text{rank}(T) = 0 &\longmapsto (Z(T), \mathcal{L}^{\vee \otimes 2}, 0). \end{aligned} \quad (4.34)$$

Remark 4.7

The well-definedness of the map follows from the explicit description of the action of Γ on the boundary.

Construction 4.8 (Construction of $\Theta_{\text{Ger},1}$). The construction of $\Theta_{\text{Ger},1}$ uses a variant of the Borcherds product. First, recall that

$$\begin{aligned} \Theta^{\text{Sp}_4}(\tau_{11}, \tau_{12}, \tau_{22}; \nu) &= \sum_{\lambda_1 \in \Lambda} \sum_{\lambda_2 \in \Lambda} e \left(\tau_{11} \frac{\lambda_{1,+}^2}{2} + \overline{\tau_{11}} \frac{\lambda_{1,-}^2}{2} \right. \\ &\quad \left. + \tau_{12} \lambda_{1,+} \lambda_{2,+} + \overline{\tau_{12}} \lambda_{1,-} \lambda_{2,-} \right. \\ &\quad \left. + \tau_{22} \frac{\lambda_{2,+}^2}{2} + \overline{\tau_{22}} \frac{\lambda_{2,-}^2}{2} \right) \\ &= \sum_{\lambda_2 \in \Lambda} \Theta^{\lambda_2}(\tau_{11}, \tau_{12}; \nu) e(\tau_{22} \lambda_{2,+}^2), \end{aligned}$$

where

$$\Theta^{\lambda_2}(\tau_{11}, \tau_{12}; \nu) = \sum_{\lambda_1 \in \Lambda} e(\tau_{11} \frac{\lambda_{1,+}^2}{2} + \overline{\tau_{11}} \frac{\lambda_{1,-}^2}{2} + \tau_{12} \lambda_{1,+} \lambda_{2,+} + \overline{\tau_{12}} \lambda_{1,-} \lambda_{2,-} + \overline{\tau_{22}} \frac{\lambda_{2,-}^2}{2}). \quad (4.35)$$

Thus

$$\theta^{\lambda_2}(\tau_{11}, \tau_{12}; \nu) := \Theta^{\lambda_2}(\tau_{11}, \tau_{12}; \nu)|_{Z(\lambda_2)} = \sum_{\lambda_1 \in \Lambda} e(\tau_{11} \frac{\lambda_{1,+}^2}{2} + \overline{\tau_{11}} \frac{\lambda_{1,-}^2}{2} + \tau_{12} \lambda_{1,+} \lambda_{2,+}). \quad (4.36)$$

Fact 4.9. *It follows from the definition that*

$$\theta^{\lambda_2}(\tau_{11}, \tau_{12}; \nu) dq_{11} \in H^1(\mathring{X}_1^{\min}, \pi_* \kappa^{-m} \otimes \omega_1^{1+n/2}).$$

We now define $\Theta^{\text{Ger},1}$. Given

$$f_m \in H^0(\mathring{X}_1^{\min}, R^1 \pi_* \kappa^{-m} \otimes \omega_1^{1-n/2}(-D)),$$

choose λ_2 with $Q(\lambda_2) = m$. Then

$$f_m(\tau_{11}, \tau_{12}) \cup \theta^{\lambda_2}(\tau_{11}, \tau_{12}; \nu) dq_{11} \in H^1(\mathring{X}_1^{\min}, \Omega_{\mathring{X}_1^{\min}}^1). \quad (4.37)$$

Definition 4.10. Carrying out the integration with respect to τ_{12} identifies the regularized integral

$$\int_{\mathring{X}_1^{\text{tor}}}^* f_m(\tau_{11}, \tau_{12}) \theta^{\lambda_2}(\tau_{11}, \tau_{12}; \nu)|_{Z(\lambda_2)} dq_{11} dq_{12}$$

with the regularized theta integral entering the Borchers lifting of a weakly holomorphic modular form $\widetilde{f}_m$ to $Z(\lambda_2)$, and hence gives rise to the corresponding Borchers product. We therefore define

$$\Theta^{\text{Ger},1}(f_m) := \text{Bor}(\widetilde{f}_m) \in K_1^{M,*}(x),$$

where Bor denotes the corresponding Borchers product and x is the generic point of $Z(\lambda_2)$.

Construction 4.11 (Construction of $\Theta^{\text{Ger},2}$). A direct construction of $\Theta^{\text{Ger},2}$, which we call the *higher Borchers product*, is difficult. We instead use the following auxiliary construction.

We briefly recall the basic geometry of the genus-2 Siegel modular variety. Let

$$X = \text{Sp}_4(\mathbf{Z}) \backslash \mathfrak{H}_2$$

denote the open Siegel modular threefold, and let

$$j : X \hookrightarrow X^{\min}$$

be its Baily–Borel, or minimal, compactification. We write ω for the Hodge line bundle on X , as well as for its canonical extension to $X^{\min}$.

By the Baily–Borel construction [BB66], the minimal compactification is obtained as the Proj of the graded ring of scalar-valued Siegel modular forms:

$$X^{\min} = \text{Proj} \left(\bigoplus_{k \geq 0} H^0(X, \omega^{\otimes k}) \right).$$

For the full-level genus-2 modular group, Igusa’s structure theorem [Igu62, Igu64] gives

$$M_{\text{even}}(\text{Sp}_4(\mathbf{Z})) = \mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}],$$

where the generators have respective weights 4, 6, 10, 12.

Here E_4 and E_6 denote the genus-2 Siegel Eisenstein series of weights 4 and 6, respectively, while χ_{10} and χ_{12} denote the standard Igusa cusp forms of weights 10 and 12. Thus

$$E_4 \in H^0(X^{\min}, \omega^{\otimes 4}), \quad E_6 \in H^0(X^{\min}, \omega^{\otimes 6}),$$

and

$$\chi_{10} \in H^0(X^{\min}, \omega^{\otimes 10}), \quad \chi_{12} \in H^0(X^{\min}, \omega^{\otimes 12}).$$

We fix the usual normalizations of these modular forms; their precise scalar normalizations will play no role below. In particular, χ_{10} and χ_{12} are cusp forms, whereas E_4 and E_6 are non-cuspidal Eisenstein series.

Consequently,

$$X^{\min} \simeq \text{Proj } \mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}].$$

After dividing all weights by 2, this may equivalently be written as

$$X^{\min} \simeq \mathbf{P}(2, 3, 5, 6),$$

where, by definition,

$$\mathbf{P}(2, 3, 5, 6) := \text{Proj } \mathbf{C}[x_2, x_3, x_5, x_6], \quad \deg(x_i) = i.$$

Equivalently, this is the weighted projective space associated with the weights $(2, 3, 5, 6)$. The passage from weights $(4, 6, 10, 12)$ to $(2, 3, 5, 6)$ is merely a regrading and does not change the underlying projective spectrum.

Moreover, the ideal of even-weight cusp forms is generated by χ_{10} and χ_{12} [Igu64]. Since the Baily–Borel boundary is defined by the homogeneous ideal of cusp forms, we have set-theoretically

$$\partial X^{\min} = X^{\min} \setminus X = V_+(\chi_{10}, \chi_{12}).$$

It follows that the open Siegel threefold is

$$X = X^{\min} \setminus V_+(\chi_{10}, \chi_{12}) = D_+(\chi_{10}) \cup D_+(\chi_{12}).$$

In particular,

$$\dim X^{\min} = 3, \quad \dim \partial X^{\min} = 1,$$

so that the Baily–Borel boundary has codimension 2 and is not a divisor.

For comparison, if

$$X \hookrightarrow X^{\text{tor}} \xrightarrow{\pi} X^{\min}$$

is a toroidal compactification, then

$$D_{\infty} := X^{\text{tor}} \setminus X$$

is a genuine boundary divisor, while π contracts its components onto the lower-dimensional strata of the Baily–Borel boundary. In what follows, however, we shall work primarily on the open variety X and its minimal compactification $X^{\min}$.

Finally, we write

$$D_{10} := V_+(\chi_{10}), \quad D_{12} := V_+(\chi_{12})$$

for the divisors on $X^{\min}$ cut out by the two cusp forms. Their restrictions to the open Siegel threefold X will be denoted by the same symbols whenever no confusion can arise.

Proposition 4.12

Assume that

$$X = \mathrm{Sp}_4(\mathbf{Z}) \backslash \mathfrak{H}_2$$

is the level-one open Siegel threefold, and let ω_X denote the Hodge line bundle. Put

$$k := 2 - \frac{n}{2} \in \mathbf{Z}.$$

Then

$$H^1(X, \omega_X^k) \simeq \frac{(\mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}][(\chi_{10}\chi_{12})^{-1}])_k}{(\mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}][\chi_{10}^{-1}])_k + (\mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}][\chi_{12}^{-1}])_k}. \quad (4.38)$$

Here the subscript k denotes the homogeneous part of weight k . Equivalently, a class in $H^1(X, \omega_X^k)$ is represented by a weight- k rational modular form having poles along both $D_{10} := \mathrm{div}(\chi_{10})$ and $D_{12} := \mathrm{div}(\chi_{12})$, modulo those representatives which extend across either D_{10} or D_{12} .

Proof. Set

$$U_{10} := D_+(\chi_{10}), \quad U_{12} := D_+(\chi_{12}).$$

Both U_{10} and U_{12} are affine, and so is their intersection

$$U_{10} \cap U_{12} = D_+(\chi_{10}\chi_{12}).$$

Consequently, the affine Cech cover

$$X = U_{10} \cup U_{12}$$

computes the coherent cohomology of ω_X^k .

For a homogeneous element f of positive weight, one has

$$H^0(D_+(f), \omega_X^k) \simeq (\mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}][f^{-1}])_k.$$

Thus the Cech complex to compute $H^1(X, \omega_X^k)$ is

$$(\mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}][\chi_{10}^{-1}])_k \oplus (\mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}][\chi_{12}^{-1}])_k \longrightarrow (\mathbf{C}[E_4, E_6, \chi_{10}, \chi_{12}][(\chi_{10}\chi_{12})^{-1}])_k,$$

with differential

$$(f, g) \longmapsto f - g.$$

Taking the cokernel gives (4.38). □

Let $Y \times Y$ denote the two products for two modular curves with common fine neat level. Fix one such product embedding

$$i : Y \times Y \hookrightarrow X,$$

and let $\mathcal{I}$ be its invertible ideal sheaf. The conormal exact sequence is

$$0 \longrightarrow \mathcal{I}/\mathcal{I}^2 \longrightarrow i^*\Omega_X^1 \longrightarrow \Omega_{Y \times Y}^1 \longrightarrow 0. \quad (4.39)$$

The Kodaira–Spencer calculation gives

$$i^*\omega_2^3 \simeq (\omega_1^2 \boxtimes \omega_1^2) \otimes \mathcal{I}/\mathcal{I}^2. \quad (4.40)$$

Since $i^*\omega_2 \simeq \omega_1 \boxtimes \omega_1$, we obtain

$$\mathcal{I}/\mathcal{I}^2 \simeq \omega_1 \boxtimes \omega_1, \quad \mathcal{I}^{-1}/\mathcal{O}_X \simeq i_*(\omega_1^{-1} \boxtimes \omega_1^{-1}). \quad (4.41)$$

For each $m \geq 1$, let

$$i_m : Y_m \hookrightarrow X$$

be the closed subscheme defined by

$$\mathcal{I}_m := \mathcal{I}^m.$$

Thus Y_m is the m -fold infinitesimal thickening of the product locus $Y \times Y$ in X , with the convention

$$Y_1 = Y \times Y, \quad i_1 = i, \quad (i_m)_* \mathcal{O}_{Y_m} = \mathcal{O}_X / \mathcal{I}^m.$$

In particular, Y_m thickens the product locus inside X , rather than thickening its two factors separately.

Set

$$\mathcal{Q}_m := \mathcal{I}_m^{-1} / \mathcal{O}_X = \mathcal{I}^{-m} / \mathcal{O}_X.$$

Since $\mathcal{I}_m$ is invertible, there is a canonical identification

$$\mathcal{Q}_m \simeq (i_m)_* i_m^* \mathcal{I}_m^{-1}.$$

Tensoring the exact sequence

$$0 \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{I}_m^{-1} \longrightarrow \mathcal{Q}_m \longrightarrow 0$$

with $\omega_2^{2-n/2}$ therefore gives the connecting morphism

$$f_m : H^0(Y_m, i_m^*(\omega_2^{2-n/2} \otimes \mathcal{I}_m^{-1})) \longrightarrow H^1(X, \omega_2^{2-n/2}). \quad (4.42)$$

For $m = 1$, this recovers

$$f_1 : H^0(Y \times Y, \omega_1^{1-n/2} \boxtimes \omega_1^{1-n/2}) \longrightarrow H^1(X, \omega_2^{2-n/2}). \quad (4.43)$$

The sheaf $\mathcal{Q}_m$ carries the natural pole-order filtration

$$0 = F_0 \mathcal{Q}_m \subset F_1 \mathcal{Q}_m \subset \cdots \subset F_m \mathcal{Q}_m = \mathcal{Q}_m, \quad F_j \mathcal{Q}_m := \mathcal{I}^{-j} / \mathcal{O}_X.$$

For $1 \leq j \leq m$, its successive quotients are

$$\begin{aligned} \mathrm{gr}_j^F \mathcal{Q}_m &= \mathcal{I}^{-j} / \mathcal{I}^{-(j-1)} \\ &\simeq i_* ((\mathcal{I} / \mathcal{I}^2)^{\otimes (-j)}) \\ &\simeq i_* (\omega_1^{-j} \boxtimes \omega_1^{-j}). \end{aligned}$$

Consequently,

$$\mathrm{gr}^F \mathcal{Q}_m \simeq \bigoplus_{j=1}^m i_* (\omega_1^{-j} \boxtimes \omega_1^{-j}). \quad (4.44)$$

After tensoring with $\omega_2^{2-n/2}$, this becomes

$$\mathrm{gr}^F (\omega_2^{2-n/2} \otimes \mathcal{Q}_m) \simeq \bigoplus_{j=1}^m i_* (\omega_1^{2-n/2-j} \boxtimes \omega_1^{2-n/2-j}). \quad (4.45)$$

Thus the successive graded pieces have parallel weights $2 - n/2 - j$, for $1 \leq j \leq m$, decreasing by one at each step, from $1 - n/2$ to $2 - n/2 - m$. This is an associated-graded decomposition; no splitting of the pole-order filtration is asserted.

Since $Y \times Y$ is affine, the successive graded pieces above have vanishing higher cohomology. Taking global sections and choosing splittings of the resulting filtration therefore yields a noncanonical

isomorphism of $\mathbb{C}$ -vector spaces

$$H^0\left(Y_m, i_m^*(\omega_2^{2-n/2} \otimes \mathcal{I}_m^{-1})\right) \simeq \bigoplus_{j=1}^m H^0\left(Y \times Y, \omega_1^{2-n/2-j} \boxtimes \omega_1^{2-n/2-j}\right).$$

Proposition 4.13

Retain the divisor convention $D_{10} = 2(Y \times Y)$ and the Čech description (4.38). Let Y_m , $\mathcal{I}_m$, and f_m be as above. The morphisms f_m , for $m \geq 1$, generate $H^1(X, \omega_2^{2-n/2})$. More precisely, the map

$$\bigoplus_{m \geq 1} H^0\left(Y_m, i_m^*(\omega_2^{2-n/2} \otimes \mathcal{I}_m^{-1})\right) \xrightarrow{\sum_m f_m} H^1(X, \omega_2^{2-n/2}) \quad (4.46)$$

is surjective.

Moreover, after choosing compatible $\mathbb{C}$ -vector-space splittings of the pole-order filtrations, one obtains a noncanonical surjection

$$\bigoplus_{j \geq 1} H^0\left(Y \times Y, \omega_1^{2-n/2-j} \boxtimes \omega_1^{2-n/2-j}\right) \longrightarrow H^1(X, \omega_2^{2-n/2}). \quad (4.47)$$

Proof. Put $k := 2 - n/2$. Recall that, up to a nonzero scalar, χ_{10} is the product of the squares of the ten even genus-two theta constants. It vanishes to order two along the decomposable locus and nowhere else on the open Siegel threefold; see [GL12, §2.3], following [Igu62]. In our divisor convention, this gives

$$D_{10} = \operatorname{div}(\chi_{10}) = 2(Y \times Y). \quad (4.48)$$

Let $\xi \in H^1(X, \omega_2^k)$. By (4.38), we may represent ξ as

$$\xi = \left[\frac{A}{\chi_{12}^a \chi_{10}^b} \right], \quad a, b \geq 1,$$

where

$$A \in \mathbb{C}[E_4, E_6, \chi_{10}, \chi_{12}]$$

is homogeneous of weight $k + 12a + 10b$. Choose

$$m := 2b.$$

Then

$$\mathcal{I}_m = \mathcal{O}_X(-m(Y \times Y)),$$

and (4.48) gives an isomorphism

$$\mathcal{I}_m^{-1} = \mathcal{O}_X(2b(Y \times Y)) = \mathcal{O}_X(bD_{10}) \simeq \omega_2^{10b}.$$

Here the isomorphism is the one sending the canonical section of $\mathcal{O}_X(bD_{10})$ to χ_{10}^b . Consequently, the natural inclusion

$$\omega_2^k \longrightarrow \omega_2^k \otimes \mathcal{I}_m^{-1}$$

identifies with multiplication by χ_{10}^b :

$$\omega_2^k \xrightarrow{\cdot \chi_{10}^b} \omega_2^{k+10b}.$$

The induced map on cohomology therefore sends ξ to

$$\chi_{10}^b \xi = \left[\frac{A}{\chi_{12}^a} \right] = 0 \quad \text{in } H^1(X, \omega_2^{k+10b}).$$

Indeed, A/χ_{12}^a is a section of ω_2^{k+10b} over $D_+(\chi_{12})$, so it represents a coboundary for the affine Čech cover

$$X = D_+(\chi_{10}) \cup D_+(\chi_{12}).$$

Thus ξ lies in the kernel of the natural map

$$H^1(X, \omega_2^k) \longrightarrow H^1(X, \omega_2^k \otimes \mathcal{I}_m^{-1}).$$

Now consider the short exact sequence

$$0 \longrightarrow \omega_2^k \longrightarrow \omega_2^k \otimes \mathcal{I}_m^{-1} \longrightarrow (i_m)_* i_m^* (\omega_2^k \otimes \mathcal{I}_m^{-1}) \longrightarrow 0.$$

Its long exact cohomology sequence contains

$$H^0(Y_m, i_m^* (\omega_2^k \otimes \mathcal{I}_m^{-1})) \xrightarrow{f_m} H^1(X, \omega_2^k) \longrightarrow H^1(X, \omega_2^k \otimes \mathcal{I}_m^{-1}).$$

By exactness, there exists

$$\eta_m \in H^0(Y_m, i_m^* (\omega_2^k \otimes \mathcal{I}_m^{-1}))$$

such that $f_m(\eta_m) = \xi$. Since ξ was arbitrary, this proves (4.46).

Finally, write

$$V_m := H^0(X, \omega_2^k \otimes \mathcal{Q}_m), \quad V_0 := 0,$$

and

$$W_j := H^0(Y \times Y, \omega_1^{k-j} \boxtimes \omega_1^{k-j}).$$

Since $Y \times Y$ is affine, the pole-order filtration described above induces short exact sequences of $\mathbb{C}$ -vector spaces

$$0 \longrightarrow V_{j-1} \longrightarrow V_j \longrightarrow W_j \longrightarrow 0.$$

Choosing a splitting at each step gives compatible noncanonical identifications

$$V_m \simeq \bigoplus_{j=1}^m W_j, \quad \varinjlim_m V_m \simeq \bigoplus_{j \geq 1} W_j.$$

Naturality of the connecting morphisms gives

$$f_{m+1}|_{V_m} = f_m.$$

Hence the maps f_m induce a surjection

$$\varinjlim_m V_m \longrightarrow H^1(X, \omega_2^k).$$

Composing with the chosen direct-sum identification proves (4.47). □

Remark 4.14

The proposition deliberately does not assert that the map

$$f_1 : H^0(Y \times Y, \dots) \longrightarrow H^1(X, \omega_2^{2-n/2})$$

is surjective. Nor does it require a Serre-vanishing statement on the open Siegel threefold.

The proposition gives the following auxiliary diagram:

$$\begin{array}{ccc}
 \bigoplus_{j \geq 1} H^0(Y \times Y, \omega_1^{2-n/2-j} \boxtimes \omega_1^{2-n/2-j}) & \longrightarrow & H^1(X, \omega_2^{2-n/2}) \\
 \downarrow (\text{Bor} \boxtimes \text{Bor})_{j=1} & & \downarrow \Theta^{\text{Ger}, 2} \\
 \text{Mero}^*(\text{Sh}_{\text{SO}(V)})^{\otimes 2} & \longrightarrow & K_2^{M,*}(\eta).
 \end{array} \tag{4.49}$$

Accordingly, the diagram relevant for the comparison with the middle term of (4.33) is

$$\begin{array}{ccc}
 \bigoplus_{j \geq 1} H^0(Y \times Y, \omega_1^{2-n/2-j} \boxtimes \omega_1^{2-n/2-j}) & \twoheadrightarrow & \text{Im}(P_2) \\
 \downarrow (\text{Bor} \boxtimes \text{Bor})_{j=1} & & \downarrow \\
 \text{Mero}^*(\text{Sh}_{\text{SO}(V)})^{\otimes 2} & \longrightarrow & \bigoplus_{x \in \text{Sh}_{\text{SO}(V)}^{(1)}} K_1^{M,*}(x).
 \end{array} \tag{4.50}$$

The right vertical arrow is the restriction of $\Theta^{\text{Ger}, 1}$ to $\text{Im}(P_2)$. By Proposition 4.13, the upper horizontal arrow is surjective.

Theorem 4.15 (The two commutative squares)

- (1) The right-hand square in (4.33) is commutative.
- (2) The square in (4.50) is commutative.

Proof. (1) The right-hand square of (4.33) can be described on generators as follows:

$$\begin{array}{ccc}
 f_n & \longmapsto & T = \begin{pmatrix} (v_1, v_1) & (v_1, v_2) \\ (v_2, v_1) & (v_2, v_2) \end{pmatrix} \\
 \downarrow & & \downarrow \\
 (\text{Bor}(f_n))_{Z(v_2)} & \longmapsto & Z(T).
 \end{array} \tag{4.51}$$

(2) The square in (4.50) can likewise be described on generators:

$$\begin{array}{ccc}
 (f, g) & \longmapsto & (f, -g) - (g, -f) \\
 \downarrow & & \downarrow \\
 \text{Bor}(f) \otimes \text{Bor}(g) & \longmapsto & \bigoplus_{Z(v)} \frac{f^{\text{ord}_{Z(v)}(g)}}{g^{\text{ord}_{Z(v)}(f)}}.
 \end{array} \tag{4.52}$$

□

4.5 The two motivic theta classes

Theorem 4.16 (The genus-two classes)

There are Serre-dual maps

$$H^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \longrightarrow \text{CH}^2(\text{Sh}_{\text{SO}(V)})_{\mathbb{C}}, \tag{4.53}$$

and

$$H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \longrightarrow \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1)_{\mathbb{C}}. \quad (4.54)$$

They give a class

$$\Theta^{\text{mot}, 0} = \Theta^a \in H^0(X_2^{\text{tor}}, \omega_2^{1+n/2}) \otimes \text{CH}^2(\text{Sh}_{\text{SO}(V)})_{\mathbb{C}} \quad (4.55)$$

such that

$$\Theta^a = \sum_{T \geq 0} Z(T) q^T, \quad (4.56)$$

and a class

$$\Theta^{\text{mot}, 1} \in H^1(X_2^{\text{tor}}, \omega_2^{1+n/2}) \otimes \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1)_{\mathbb{C}}. \quad (4.57)$$

Proof. First, the Serre duality pairing

$$\langle , \rangle : H^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \times H^0(X_2^{\text{tor}}, \omega_2^{1+n/2}) \longrightarrow \mathbb{C} \quad (4.58)$$

can be expressed in terms of local-cohomology representatives as follows. For any

$$f \in H^0(X_2^{\text{tor}}, \omega_2^{1+n/2}) \quad (4.59)$$

and any $x \in X_0^{\text{min}}$, write the q -expansion of f near $\pi^{-1}(x)$ as

$$f = \sum_{T \geq 0} a_T q^T.$$

For any

$$g = \sum_{T \leq 0} b_T q^T \in H_{X_0^{\text{tor}}}^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)), \quad (4.60)$$

the Serre duality pairing is given by

$$\langle f, g \rangle := \sum_{T \leq 0} b_T a_T.$$

Next, write the q -expansion of Θ^a as

$$\Theta^a = \sum_{T \geq 0} a_T q^T.$$

Then

$$\langle \Theta^a, q^{-T} \rangle = a_T = \Theta^{\text{Ger}, 0}(q^{-T}) = Z(T).$$

□

Corollary 4.17 (Kudla's modularity conjecture)

The arithmetic theta series $\sum_{T \geq 0} Z(T) q^T$ is a Siegel modular form of weight $1 + n/2$. This gives a new proof of Kudla's modularity conjecture in genus two.

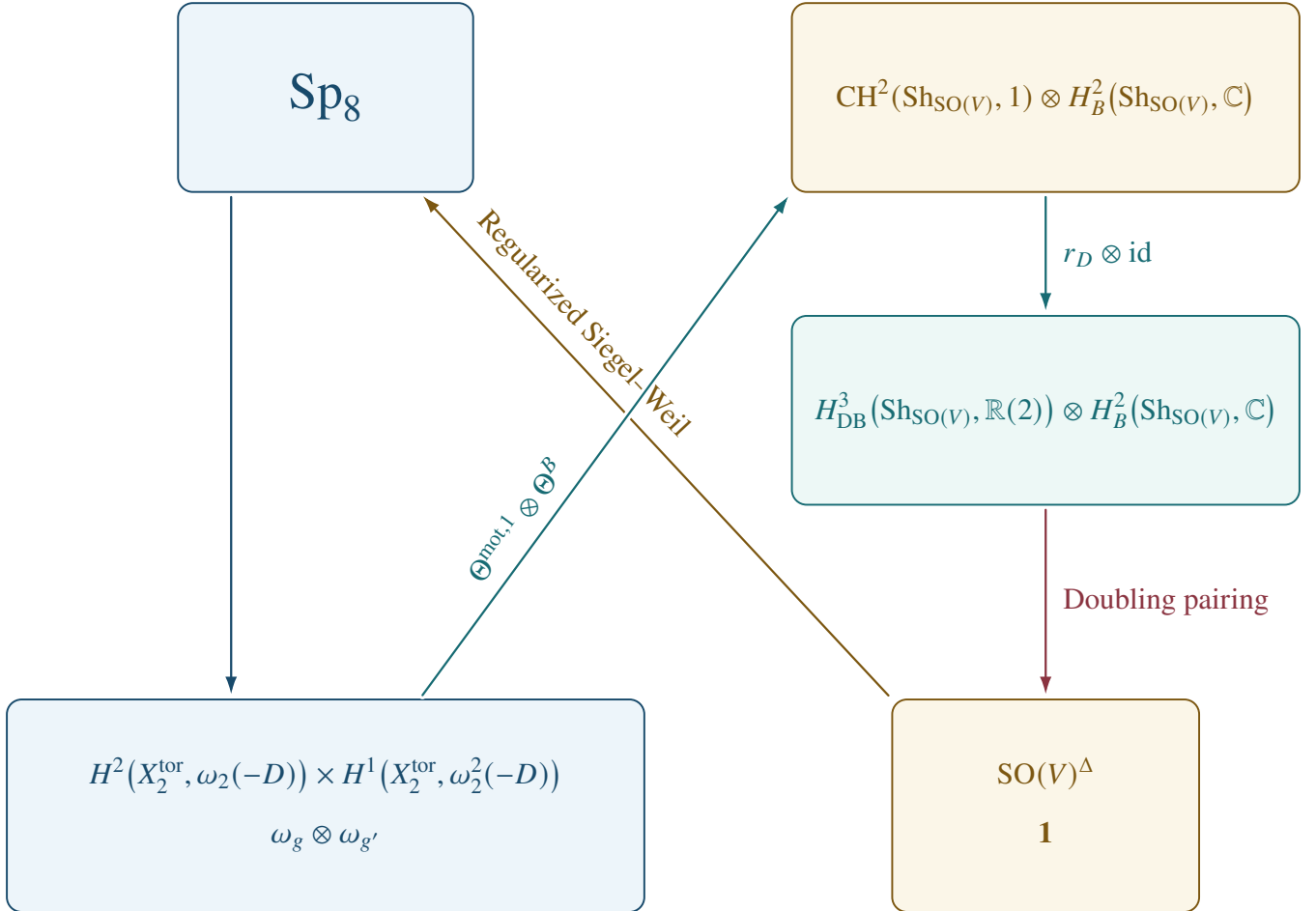
Remark 4.18

This proof provides a uniform, conceptual approach to the modularity statement.

Part 2

Relation to L -values

Motivic doubling seesaw



5 Serre duality via the Dolbeault resolution

5.1 The genus-one residue pairing

Continue with the notation of Section 3. Write $X = X_\Gamma$, $Y = Y_\Gamma$, and $\omega_1 = \omega$, and fix one cusp, written as ∞ . Proposition 3.1 gives the cokernel description of the compactified coherent cohomology. We now record its Serre-dual residue pairing.

Serre duality gives

$$H^1(X, \omega_1^{1-n/2}(-D)) \times H^0(X, \omega_1^{n/2-1}(D) \otimes \Omega_X^1) \longrightarrow \mathbb{C}. \quad (5.1)$$

In the Dolbeault realization, if η is a $(0, 1)$ -form with values in the first bundle and w is a $(1, 0)$ -form with values in the dual bundle, then

$$\langle \eta, w \rangle = \int_X \eta \wedge w. \quad (5.2)$$

Remark 5.1

This pairing agrees with the local-cohomology pairing. Around ∞ , write

$$g = \sum_{n \geq 0} a_n q^{-n} \in H_D^1(X) \simeq \mathbb{C}[q^{-1}], \quad f = \sum_{n \geq 0} b_n q^n dq \in H^0(X, \Omega_X^1). \quad (5.3)$$

The residue theorem gives

$$\langle f, g \rangle = \sum_{n \geq 0} a_n b_n. \quad (5.4)$$

5.2 Serre duality for the Siegel threefold

For

$$\tau = \begin{pmatrix} \tau_{11} & \tau_{12} \\ \tau_{12} & \tau_{22} \end{pmatrix} \in \mathbb{H}_2, \quad \text{Im}(\tau) > 0,$$

we consider the pairing

$$H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \times H^1(X_2^{\text{tor}}, \Lambda^3 \Omega_{X_2^{\text{tor}}}^1 \otimes \omega_2^{n/2-2}(D)) \longrightarrow \mathbb{C}. \quad (5.5)$$

If η is a $(0, 2)$ -form with values in $\omega_2^{2-n/2}(-D)$ and w is a $(3, 1)$ -form with values in $\omega_2^{n/2-2}(D)$, then

$$\langle \eta, w \rangle = \int_{X_2^{\text{tor}}} \eta \wedge w. \quad (5.6)$$

5.3 Local-cohomology representatives

The cohomology class is represented through

$$H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \simeq \frac{\ker P_1}{\text{Im } P_2}. \quad (5.7)$$

Write a representative as

$$\eta = \sum_{m \geq 0} f_m q_{22}^{-m}. \quad (5.8)$$

For $\eta \in \ker P_1$, we use the exact sequence

$$0 \longrightarrow H_{X_1^{\text{tor}}}^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \longrightarrow H_{X_1^{\text{tor}}}^2(X_2^{\text{tor}} \setminus X_0^{\text{tor}}, \omega_2^{2-n/2}(-D)) \xrightarrow{P_1} H_{X_0^{\text{tor}}}^3(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)). \quad (5.9)$$

Choose a lift η' through (5.9) and write

$$\eta' = \sum_{m \geq 0} f'_m q_{22}^{-m}. \quad (5.10)$$

The two growth conditions are: f'_m has moderate growth along q_{11} , whereas f_m has exponential growth along q_{11} . Thus

$$\boxed{\eta' \text{ is a lift of } \eta \text{ for } \eta \in \ker P_1.} \quad (5.11)$$

6 The Beilinson regulator and the joint Kudla–Millson–Deligne–Beilinson classes

6.1 The map to the higher Chow group

Recall the map constructed in Theorem 4.16:

$$\Theta : H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \longrightarrow \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1) = \frac{\ker \left(\bigoplus_{x \in \text{Sh}_{\text{SO}(V)}^{(1)}} K_1^{M,*}(x) \xrightarrow{\text{div}^*} Z^{2,*}(\text{Sh}_{\text{SO}(V)}) \right)}{\text{Im} \left(K_2^{M,*}(\eta) \xrightarrow{T^*} \bigoplus_{x \in \text{Sh}_{\text{SO}(V)}^{(1)}} K_1^{M,*}(x) \right)}. \quad (6.1)$$

Put

$$\begin{aligned} V &= H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)), \\ W &= \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1). \end{aligned} \quad (6.2)$$

The identity $\text{Hom}_{\mathbb{C}}(V, W) \simeq V^{\vee} \otimes W$, together with Serre duality, identifies Θ with the tensor

$$\Theta^{\text{mot}, 1} \in H^1(X_2^{\text{tor}}, \Lambda^3 \Omega_{X_2^{\text{tor}}}^1 \otimes \omega_2^{n/2-2}(D)) \otimes \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1) \simeq H^1(X_2^{\text{tor}}, \omega_2^{1+n/2}) \otimes \text{CH}^2(\text{Sh}_{\text{SO}(V)}, 1). \quad (6.3)$$

Remark 6.1

Since the Serre duality pairing is perfect, this tensor is characterized by

$$\langle \Theta^{\text{mot}, 1}, \eta \rangle = \Theta(\eta) \quad (6.4)$$

for every $\eta \in H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D))$.

6.2 The Beilinson regulator

We recall the following description of Deligne–Beilinson cohomology.

Proposition 6.2 ([BGCLRJ24a, Theorem 2.25])

The real Deligne–Beilinson cohomology group $H_D^3(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2))$ admits the canonical description

$$H_{\text{DB}}^3(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2)) \simeq \frac{\{(S, T) \in \mathcal{D}_{\text{si}, \mathbb{R}(1)}^2(\overline{\text{Sh}_{\text{SO}(V)}}) \oplus F^2 \mathcal{D}_{\text{si}}^3(\overline{\text{Sh}_{\text{SO}(V)}}) \mid dT = 0, dS = \frac{1}{2}(T - \bar{T})\}}{\{d(\tilde{S}, \tilde{T})\}},$$

where $d(\tilde{S}, \tilde{T}) = (d\tilde{S} - \pi_1(\tilde{T}), d\tilde{T})$.

Remark 6.3

Here $\mathcal{D}_{si}^*(\mathrm{Sh}_{\mathrm{SO}(V)})$ denotes the space of tempered currents; see [BGCLRJ24a, Definition 2.19].

Recall that the Beilinson regulator

$$r_D : \mathrm{CH}^2(\mathrm{Sh}_{\mathrm{SO}(V)}, 1)_{\mathbb{Q}} \cong H_M^3(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{Q}(2)) \longrightarrow H_D^3(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{R}(2)), \quad (6.5)$$

has the explicit representative

$$\bigoplus_{z \in Z^1(\mathrm{Sh}_{\mathrm{SO}(V)})} f_z \longmapsto (S = \sum_z (\log |f_z|) \delta_z, T). \quad (6.6)$$

Remark 6.4

In this description, T is determined by S , so we do not write it explicitly.

Definition 6.5. Applying the Beilinson regulator to $\Theta^{\mathrm{mot},1}$ gives

$$\Theta^{\mathrm{DB},1} := r_D(\Theta^{\mathrm{mot},1}) \in H^1(X_2^{\mathrm{tor}}, \Lambda^3 \Omega_{X_2^{\mathrm{tor}}}^1 \otimes \omega_2^{n/2-2}(D)) \otimes H_{\mathrm{DB}}^3(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{R}(2)). \quad (6.7)$$

Remark 6.6

Our next goal is to describe $\Theta^{\mathrm{DB},1}$ explicitly.

6.3 The Fock model

Notation 6.7. • **Symplectic Lie algebra.** Let W be the standard symplectic space with basis $e_1, \dots, e_r, f_1, \dots, f_r$, and put

$$w'_j = e_j + i f_j, \quad w''_j = e_j - i f_j, \quad W_{\mathbb{C}} = W' \oplus W''.$$

We use the standard identification

$$\begin{aligned} \mathrm{Sym}^2(W_{\mathbb{C}}) &\xrightarrow{\sim} \mathfrak{sp}(W)_{\mathbb{C}}, \\ x \circ y &:= x \otimes y + y \otimes x \longmapsto (z \longmapsto \langle x, z \rangle y + \langle y, z \rangle x). \end{aligned} \quad (6.8)$$

For $1 \leq i \leq j \leq r$, define

$$L_{ij} := \frac{1}{4i} w''_i \circ w''_j \in \mathfrak{p}_r^-. \quad (6.9)$$

Thus $L_{ij} = L_{ji}$, and

$$\mathfrak{sp}(W)_{\mathbb{C}} = \mathfrak{k}_{r,\mathbb{C}} \oplus \mathfrak{p}_r^+ \oplus \mathfrak{p}_r^- \quad (6.10)$$

$$\mathfrak{p}_r^- = \mathrm{Span}_{\mathbb{C}} \{L_{ij} : 1 \leq i \leq j \leq r\}. \quad (6.11)$$

- **Orthogonal Lie algebra.** Let $V = V_+ \oplus V_-$ have signature $(n, 2)$, with bases $v_1, \dots, v_n$ of V_+ and v_{n+1}, v_{n+2} of V_- . We use the identification

$$\begin{aligned} \wedge^2(V_{\mathbb{C}}) &\xrightarrow{\sim} \mathfrak{so}(V)_{\mathbb{C}}, \\ v \wedge w &\longmapsto (x \longmapsto (v, x)w - (w, x)v). \end{aligned} \quad (6.12)$$

For $1 \leq i < j \leq n+2$, define

$$X_{ij} := v_i \wedge v_j \in \wedge^2(V_{\mathbb{C}}) \simeq \mathfrak{so}(V)_{\mathbb{C}}. \quad (6.13)$$

Equivalently,

$$X_{ij}(x) = (v_i, x)v_j - (v_j, x)v_i. \quad (6.14)$$

In particular, for $1 \leq \alpha \leq n$, put

$$Y_\alpha^+ := X_{\alpha,n+1} + iX_{\alpha,n+2}, \quad Y_\alpha^- := X_{\alpha,n+1} - iX_{\alpha,n+2}. \quad (6.15)$$

Then

$$\mathfrak{p}_V^\pm = \text{Span}_{\mathbb{C}} \{Y_\alpha^\pm : 1 \leq \alpha \leq n\}. \quad (6.16)$$

We denote the dual bases of $(\mathfrak{p}_V^+)^*$ and $(\mathfrak{p}_V^-)^*$ by $d\tau_\alpha$ and $d\bar{\tau}_\alpha$, respectively.

Definition 6.8 (Fock model). For the symplectic space

$$\mathbf{W} = V \otimes W,$$

use the polarization

$$\begin{aligned} \mathbf{W}' &= (V_+ \otimes W') \oplus (V_- \otimes W''), \\ \mathbf{W}'' &= (V_+ \otimes W'') \oplus (V_- \otimes W'). \end{aligned} \quad (6.17)$$

Set

$$z_{\alpha j} := v_\alpha \otimes w'_j \quad \text{for } \alpha = 1, \dots, n \text{ and } j = 1, \dots, r, \quad (6.18)$$

$$z_{\mu j} := v_\mu \otimes w''_j \quad \text{for } \mu = n+1, n+2 \text{ and } j = 1, \dots, r, \quad (6.19)$$

The Fock model is the polynomial representation

$$\mathcal{P}_r := \text{Sym}^\bullet(\mathbf{W}') = \mathbb{C}[z_{\alpha j}, z_{\mu j}], \quad \begin{matrix} 1 \leq \alpha \leq n, \\ 1 \leq j \leq r \end{matrix}, \quad \begin{matrix} \mu \in \{n+1, n+2\}, \\ 1 \leq j \leq r \end{matrix}. \quad (6.20)$$

The Lie algebra action is given by

$$\begin{aligned} \rho(v_\alpha \otimes w'_j) &= z_{\alpha j}, & \rho(v_\alpha \otimes w''_j) &= -4\pi \frac{\partial}{\partial z_{\alpha j}}, \\ \rho(v_\mu \otimes w'_j) &= -4\pi \frac{\partial}{\partial z_{\mu j}}, & \rho(v_\mu \otimes w''_j) &= z_{\mu j}. \end{aligned} \quad (6.21)$$

Notation 6.9. • To simplify the notation, write

$$\begin{aligned} \lambda_j &:= z_{n+1,j} + iz_{n+2,j}, & \bar{\lambda}_j &:= z_{n+1,j} - iz_{n+2,j}, \\ Z_j &:= \sum_{\alpha=1}^n z_{\alpha j} d\tau_\alpha, & \bar{Z}_j &:= \sum_{\alpha=1}^n z_{\alpha j} d\bar{\tau}_\alpha. \\ \omega &:= \sum_{\alpha=1}^n d\tau_\alpha \wedge d\bar{\tau}_\alpha. \end{aligned} \quad (6.22)$$

- Let e generate the one-dimensional K_r -type $\det^{-(n+2)/2}$, and set

$$M = \mathbb{C}[z_{\alpha j}, \lambda_j, \bar{\lambda}_j] \otimes \mathbb{C}e. \quad (6.23)$$

The symplectic differential operators act by

$$\rho(L_{jk})(f \otimes e) = \left(-2\pi \sum_{\alpha=1}^n \frac{\partial^2}{\partial z_{\alpha j} \partial z_{\alpha k}} + \frac{\lambda_j \bar{\lambda}_k + \bar{\lambda}_j \lambda_k}{16\pi} \right) f \otimes e. \quad (6.24)$$

On the orthogonal side, write

$$\partial_V = \sum_{j=1}^r \partial_V^{(j)}, \quad \bar{\partial}_V = \sum_{j=1}^r \bar{\partial}_V^{(j)},$$

where

$$\begin{aligned}\partial_V^{(j)}(f \otimes e \otimes \eta) &= \sum_{\alpha=1}^n \left(\frac{z_{\alpha j} \lambda_j}{4\pi} - 8\pi \frac{\partial^2}{\partial z_{\alpha j} \partial \bar{\lambda}_j} \right) f \otimes e \otimes d\tau_\alpha \wedge \eta, \\ \bar{\partial}_V^{(j)}(f \otimes e \otimes \eta) &= \sum_{\alpha=1}^n \left(\frac{z_{\alpha j} \bar{\lambda}_j}{4\pi} - 8\pi \frac{\partial^2}{\partial z_{\alpha j} \partial \lambda_j} \right) f \otimes e \otimes d\bar{\tau}_\alpha \wedge \eta.\end{aligned}\tag{6.25}$$

• For

$$\phi \in \text{Hom}_{\mathfrak{k}_{r,\mathbb{C}}}(\wedge^a \mathfrak{p}_r^-, M),$$

the relative Lie algebra differential is

$$\begin{aligned}(d_{\text{Sp}}\phi)(X_1 \wedge \cdots \wedge X_{a+1}) \\ = \sum_{\nu=1}^{a+1} (-1)^{\nu+1} \rho(X_\nu) \phi(X_1 \wedge \cdots \wedge \widehat{X_\nu} \wedge \cdots \wedge X_{a+1}).\end{aligned}\tag{6.26}$$

Remark 6.10 (The case $r = 2$)

We now specialize the preceding conventions to $r = 2$. In this case, W has dimension four, and

$$\mathfrak{sp}_{4,\mathbb{C}} = \mathfrak{sp}(W)_{\mathbb{C}} = \mathfrak{k}_{2,\mathbb{C}} \oplus \mathfrak{p}_2^+ \oplus \mathfrak{p}_2^-, \tag{6.27}$$

where

$$\begin{aligned}\mathfrak{k}_{2,\mathbb{C}} &= \text{Span}_{\mathbb{C}}\{H_{11}, H_{12}, H_{21}, H_{22}\}, \\ \mathfrak{p}_2^+ &= \text{Span}_{\mathbb{C}}\{R_{11}, R_{12}, R_{22}\}, \\ \mathfrak{p}_2^- &= \text{Span}_{\mathbb{C}}\{L_{11}, L_{12}, L_{22}\}.\end{aligned}\tag{6.28}$$

The real and complex Fock modules are

$$\begin{aligned}M_{\mathbb{R}} &:= \mathbb{R}[z_{\alpha j}, \lambda_j, \bar{\lambda}_j \mid 1 \leq \alpha \leq n, j = 1, 2] \otimes_{\mathbb{R}} \mathbb{R}\mathbf{e}, \\ M &:= M_{\mathbb{C}} := M_{\mathbb{R}} \otimes_{\mathbb{R}} \mathbb{C} \\ &= \mathbb{C}[z_{\alpha j}, \lambda_j, \bar{\lambda}_j \mid 1 \leq \alpha \leq n, j = 1, 2] \otimes_{\mathbb{C}} \mathbb{C}\mathbf{e},\end{aligned}\tag{6.29}$$

where

$$\lambda_j = z_{n+1,j} + iz_{n+2,j}, \quad \bar{\lambda}_j = z_{n+1,j} - iz_{n+2,j}, \quad j = 1, 2. \tag{6.30}$$

The following operators are first defined on $M_{\mathbb{R}}$ and then extended $\mathbb{C}$ -linearly to M . For $1 \leq j \leq k \leq 2$, one has

$$\rho(L_{jk})(f \otimes \mathbf{e}) = \left(-2\pi \sum_{\alpha=1}^n \frac{\partial^2}{\partial z_{\alpha j} \partial z_{\alpha k}} + \frac{\lambda_j \bar{\lambda}_k + \bar{\lambda}_j \lambda_k}{16\pi} \right) f \otimes \mathbf{e}, \tag{6.31}$$

$$\rho(R_{jk})(f \otimes \mathbf{e}) = \left(\frac{1}{8\pi} \sum_{\alpha=1}^n z_{\alpha j} z_{\alpha k} - 4\pi \left(\frac{\partial^2}{\partial \lambda_j \partial \bar{\lambda}_k} + \frac{\partial^2}{\partial \bar{\lambda}_j \partial \lambda_k} \right) \right) f \otimes \mathbf{e}. \tag{6.32}$$

For $1 \leq j, k \leq 2$, the action of H_{jk} is

$$\begin{aligned}\rho(H_{jk})(f \otimes \mathbf{e}) &= \left[\frac{n-2}{2} \delta_{jk} + \frac{1}{2} \sum_{\alpha=1}^n \left(z_{\alpha j} \frac{\partial}{\partial z_{\alpha k}} + z_{\alpha k} \frac{\partial}{\partial z_{\alpha j}} \right) \right. \\ &\quad \left. - \frac{1}{2} \left(\lambda_j \frac{\partial}{\partial \lambda_k} + \bar{\lambda}_j \frac{\partial}{\partial \bar{\lambda}_k} + \lambda_k \frac{\partial}{\partial \lambda_j} + \bar{\lambda}_k \frac{\partial}{\partial \bar{\lambda}_j} \right) \right] f \otimes \mathbf{e}.\end{aligned}\tag{6.33}$$

On the orthogonal side, the relative Lie algebra differentials decompose as

$$\partial_V = \partial_V^{(1)} + \partial_V^{(2)}, \quad \bar{\partial}_V = \bar{\partial}_V^{(1)} + \bar{\partial}_V^{(2)}. \quad (6.34)$$

More precisely, for $j = 1, 2$,

$$\partial_V^{(j)}(f \otimes \mathbf{e} \otimes \omega) = \sum_{\alpha=1}^n \left(\frac{1}{4\pi} z_{\alpha j} \lambda_j - 8\pi \frac{\partial^2}{\partial z_{\alpha j} \partial \bar{\lambda}_j} \right) f \otimes \mathbf{e} \otimes d\tau_\alpha \wedge \omega, \quad (6.35)$$

$$\bar{\partial}_V^{(j)}(f \otimes \mathbf{e} \otimes \omega) = \sum_{\alpha=1}^n \left(\frac{1}{4\pi} z_{\alpha j} \bar{\lambda}_j - 8\pi \frac{\partial^2}{\partial z_{\alpha j} \partial \lambda_j} \right) f \otimes \mathbf{e} \otimes d\bar{\tau}_\alpha \wedge \omega. \quad (6.36)$$

For later use, the relevant exterior powers of $\mathfrak{p}_2^-$ have the ordered bases

$$\begin{aligned} \mathfrak{p}_2^- &= \text{Span}_{\mathbb{C}}\{L_{11}, L_{12}, L_{22}\}, \\ \wedge^2 \mathfrak{p}_2^- &= \text{Span}_{\mathbb{C}}\{L_{11} \wedge L_{12}, L_{11} \wedge L_{22}, L_{12} \wedge L_{22}\}, \\ \wedge^3 \mathfrak{p}_2^- &= \mathbb{C}(L_{11} \wedge L_{12} \wedge L_{22}). \end{aligned} \quad (6.37)$$

In particular,

$$\dim_{\mathbb{C}} \mathfrak{p}_2^- = 3, \quad \wedge^4 \mathfrak{p}_2^- = 0.$$

Consequently, the components appearing in the joint class are determined as follows:

- a map defined on $\mathfrak{p}_2^-$ is determined by its values on L_{11}, L_{12}, L_{22} ;
- a map defined on $\wedge^2 \mathfrak{p}_2^-$ is determined by its values on

$$L_{11} \wedge L_{12}, \quad L_{11} \wedge L_{22}, \quad L_{12} \wedge L_{22};$$

- a map defined on $\wedge^3 \mathfrak{p}_2^-$ is determined by its value on

$$L_{11} \wedge L_{12} \wedge L_{22}.$$

6.4 Joint Deligne–Beilinson relative Lie algebra cohomology

Let

$$\mathfrak{g}_{V,\mathbb{C}} = \mathfrak{k}_{V,\mathbb{C}} \oplus \mathfrak{p}_V^+ \oplus \mathfrak{p}_V^-$$

be the Hodge decomposition on the orthogonal side, and let

$$\mathfrak{q}_r = \mathfrak{k}_{r,\mathbb{C}} \oplus \mathfrak{p}_r^- \subset \mathfrak{sp}(W)_{\mathbb{C}}$$

be the parabolic subalgebra used on the symplectic side. Let M be the Fock module introduced above. The actions of $\mathfrak{g}_V$ and $\mathfrak{q}_r$ on M commute.

We use a for the symplectic relative Lie algebra degree and b for the Deligne–Beilinson degree.

6.4.1 The orthogonal complex

Let $Y_\alpha^\pm$ be the bases of $\mathfrak{p}_V^\pm$, and let $d\tau_\alpha$ and $d\bar{\tau}_\alpha$ be the corresponding dual bases. Put

$$A_{\mathbb{C}}^{s,t}(M) := (M \otimes \wedge^s (\mathfrak{p}_V^+)^* \otimes \wedge^t (\mathfrak{p}_V^-)^*)^{\mathfrak{k}_{V,\mathbb{C}}}. \quad (6.38)$$

The two components of the relative Lie algebra differential are

$$\partial_V(m \otimes \omega) := \sum_{\alpha=1}^n Y_\alpha^+(m) \otimes d\tau_\alpha \wedge \omega, \quad (6.39)$$

$$\bar{\partial}_V(m \otimes \omega) := \sum_{\alpha=1}^n Y_\alpha^-(m) \otimes d\bar{\tau}_\alpha \wedge \omega. \quad (6.40)$$

Thus

$$\partial_V : A_{\mathbb{C}}^{s,t}(M) \longrightarrow A_{\mathbb{C}}^{s+1,t}(M), \quad \bar{\partial}_V : A_{\mathbb{C}}^{s,t}(M) \longrightarrow A_{\mathbb{C}}^{s,t+1}(M).$$

Let c denote complex conjugation. It acts on M and interchanges $d\tau_\alpha$ with $d\bar{\tau}_\alpha$. We write

$$A_{\mathbb{R}}^m(M(r-1)) := (M \otimes \wedge^m((\mathfrak{p}_V^+)^* \oplus (\mathfrak{p}_V^-)^*))^{\mathfrak{k}_V, c=(-1)^{r-1}}. \quad (6.41)$$

The Tate twist is encoded by the $(-1)^{r-1}$ -eigenspace for c . We also put

$$F^r A_{\mathbb{C}}^b(M) := \bigoplus_{\substack{s+t=b \\ s \geq r}} A_{\mathbb{C}}^{s,t}(M). \quad (6.42)$$

Definition 6.11. The Deligne–Beilinson relative Lie algebra complex is the mapping fiber

$$C_{\text{DB}}^\bullet(M(r)) := \text{Fib}\left(F^r A_{\mathbb{C}}^\bullet(M) \xrightarrow{\text{pr}_{\mathbb{R}}} A_{\mathbb{R}}^\bullet(M(r-1))\right), \quad (6.43)$$

where

$$\text{Fib}(f) := \text{Cone}(f)[-1].$$

Equivalently,

$$C_{\text{DB}}^0(M(r)) := 0, \quad C_{\text{DB}}^b(M(r)) := A_{\mathbb{R}}^{b-1}(M(r-1)) \oplus F^r A_{\mathbb{C}}^b(M) \quad (b \geq 1). \quad (6.44)$$

For

$$\eta \in A_{\mathbb{R}}^{b-1}(M(r-1)), \quad \omega \in F^r A_{\mathbb{C}}^b(M),$$

the real projection is

$$\text{pr}_{\mathbb{R}}(\omega) := \frac{1}{2} \left(\omega + (-1)^{r-1} \bar{\omega} \right). \quad (6.45)$$

With the mapping-fiber convention above, the differential is

$$d_D(\eta, \omega) := \left(-(\partial_V + \bar{\partial}_V)\eta - \text{pr}_{\mathbb{R}}(\omega), (\partial_V + \bar{\partial}_V)\omega \right). \quad (6.46)$$

Its cohomology is denoted by

$$H_{\text{DB}}^b(\mathfrak{g}_V, \mathfrak{k}_V; M(r)) := H^b(C_{\text{DB}}^\bullet(M(r))).$$

6.4.2 The symplectic differential

For a $\mathfrak{q}_r$ -module N , put

$$C_{\text{Sp}}^a(N) := \text{Hom}_{\mathfrak{k}_{r,\mathbb{C}}}(\wedge^a \mathfrak{p}_r^-, N). \quad (6.47)$$

Since the Lie bracket on

$$\mathfrak{q}_r / \mathfrak{k}_{r,\mathbb{C}} \simeq \mathfrak{p}_r^-$$

is zero, the relative Lie algebra differential is

$$\begin{aligned} (d_{\text{Sp}} f)(X_1 \wedge \cdots \wedge X_{a+1}) \\ = \sum_{\nu=1}^{a+1} (-1)^{\nu+1} X_\nu \cdot f(X_1 \wedge \cdots \wedge \widehat{X_\nu} \wedge \cdots \wedge X_{a+1}). \end{aligned} \quad (6.48)$$

6.4.3 The joint double complex

Definition 6.12. The joint Deligne–Beilinson relative Lie algebra complex is the double complex

$$C_{\text{joint}}^{a,b}(M(r)) := \text{Hom}_{\mathfrak{k}_{r,\mathbb{C}}}(\wedge^a \mathfrak{p}_r^-, C_{\text{DB}}^b(M(r))). \quad (6.49)$$

Its two differentials are

$$d_{\text{Sp}} : C_{\text{joint}}^{a,b} \longrightarrow C_{\text{joint}}^{a+1,b}, \quad d_D : C_{\text{joint}}^{a,b} \longrightarrow C_{\text{joint}}^{a,b+1}. \quad (6.50)$$

The associated total complex is

$$C_{\text{Tot}}^m(M(r)) := \bigoplus_{a+b=m} C_{\text{joint}}^{a,b}(M(r)). \quad (6.51)$$

On $C_{\text{joint}}^{a,b}$, we use the total differential

$$d_{\text{Tot}} := d_{\text{Sp}} + (-1)^a d_D. \quad (6.52)$$

We denote its cohomology by

$$H_{\text{joint,DB}}^m(M(r)) := H^m(C_{\text{Tot}}^\bullet(M(r))). \quad (6.53)$$

Remark 6.13

The two relative Lie algebra constructions may be performed in either order, since the $\mathfrak{q}_r$ - and $\mathfrak{g}_V$ -actions on M commute.

6.4.4 The degree-four staircase

For the genus-two construction, $r = 2$ and

$$\dim_{\mathbb{C}} \mathfrak{p}_2^- = 3, \quad \wedge^4 \mathfrak{p}_2^- = 0.$$

Hence a joint cochain of total degree four has the form

$$\Phi = \phi^{(04)} + \phi^{(13)} + \phi^{(22)} + \phi^{(31)}, \quad (6.54)$$

where

$$\phi^{(a,4-a)} \in C_{\text{joint}}^{a,4-a}(M(2)).$$

With the convention in (6.52), the condition $d_{\text{Tot}} \Phi = 0$ is equivalent to

$$\begin{aligned} d_D \phi^{(04)} &= 0, & d_D \phi^{(13)} &= d_{\text{Sp}} \phi^{(04)}, \\ d_D \phi^{(22)} &= -d_{\text{Sp}} \phi^{(13)}, & d_D \phi^{(31)} &= d_{\text{Sp}} \phi^{(22)}, \\ d_{\text{Sp}} \phi^{(31)} &= 0. \end{aligned} \quad (6.55)$$

After replacing $\phi^{(22)}$ and $\phi^{(31)}$ by their negatives, these equations take the unsigned staircase form used below:

$$\begin{aligned} d_D \phi^{(04)} &= 0, & d_D \phi^{(13)} &= d_{\text{Sp}} \phi^{(04)}, \\ d_D \phi^{(22)} &= d_{\text{Sp}} \phi^{(13)}, & d_D \phi^{(31)} &= d_{\text{Sp}} \phi^{(22)}, \\ d_{\text{Sp}} \phi^{(31)} &= 0. \end{aligned} \quad (6.56)$$

6.5 The joint Kudla–Millson–Deligne–Beilinson class for $r = 2$

We now construct the total-degree-four class in the joint Deligne–Beilinson relative Lie algebra complex. Put

$$\mathfrak{k} := \mathfrak{k}_{2,\mathbb{C}}, \quad \mathfrak{p}_2^- = \text{Span}_{\mathbb{C}}\{L_{11}, L_{12}, L_{22}\}.$$

We use the ordered bases

$$\begin{aligned} \mathcal{B}_1 &:= (L_{11}, L_{12}, L_{22}), \\ \mathcal{B}_2 &:= (L_{11} \wedge L_{12}, L_{11} \wedge L_{22}, L_{12} \wedge L_{22}), \\ \mathcal{B}_3 &:= (L_{11} \wedge L_{12} \wedge L_{22}). \end{aligned} \quad (6.57)$$

Throughout this subsection, the common tensor factor e in the Fock module is suppressed. Since $r = 2$, the real components lie in the (-1) -eigenspace for complex conjugation.

6.5.1 The rank-one building blocks

For $j = 1, 2$, recall that

$$\begin{aligned}\lambda_j &:= z_{n+1,j} + iz_{n+2,j}, & \bar{\lambda}_j &:= z_{n+1,j} - iz_{n+2,j}, \\ Z_j &:= \sum_{\alpha=1}^n z_{\alpha j} d\tau_{\alpha}, & \bar{Z}_j &:= \sum_{\alpha=1}^n z_{\alpha j} d\bar{\tau}_{\alpha}.\end{aligned}\tag{6.58}$$

The two rank-one Kudla–Millson forms in the j -th Fock variable are

$$\phi_{\text{KM}}^{\text{hol},(j)} := \frac{1}{4\pi} \lambda_j Z_j, \quad \phi_{\text{KM}}^{\text{cyc},(j)} := -\frac{1}{2\pi} Z_j \wedge \bar{Z}_j.\tag{6.59}$$

They provide the basic factors in the genus-two construction.

6.5.2 The four components

Write

$$\Phi = \phi_2^{(0,4)} + \phi_2^{(1,3)} + \phi_2^{(2,2)} + \phi_2^{(3,1)}.\tag{6.60}$$

We organize its components as

$$\begin{aligned}\phi_2^{(0,4)} &= \omega_{22}, & \phi_2^{(1,3)} &= (\kappa_2, \omega_{21}), \\ \phi_2^{(2,2)} &= (\eta_2, \omega_2), & \phi_2^{(3,1)} &= \eta_3.\end{aligned}\tag{6.61}$$

Here κ_2 and η_2 are the real components in the Deligne–Beilinson complex, while ω_{21} , ω_2 , and ω_{22} are the indicated Hodge components. All components not specified below are zero. The displayed values on the ordered bases in (6.57) specify the cochains; their $\mathfrak{f}$ -equivariance is part of the verification.

The $(0, 4)$ -component. The top Hodge component is the product of the two rank-one cycle forms:

$$\begin{aligned}\omega_{22} &:= \phi_{\text{KM}}^{\text{cyc},(1)} \wedge \phi_{\text{KM}}^{\text{cyc},(2)} \\ &= \frac{1}{4\pi^2} Z_1 \wedge \bar{Z}_1 \wedge Z_2 \wedge \bar{Z}_2.\end{aligned}\tag{6.62}$$

The $(1, 3)$ -component. To simplify the notation, we denote the real degree-two component $\phi_2^{(1,3),2}$ by κ_2 . It is the polarized cycle-class cochain determined by

$$\begin{aligned}\kappa_2(L_{11}) &:= -\frac{1}{2\pi} Z_2 \wedge \bar{Z}_2, \\ \kappa_2(L_{12}) &:= \frac{1}{4\pi} (Z_1 \wedge \bar{Z}_2 + Z_2 \wedge \bar{Z}_1), \\ \kappa_2(L_{22}) &:= -\frac{1}{2\pi} Z_1 \wedge \bar{Z}_1.\end{aligned}\tag{6.63}$$

In particular,

$$\kappa_2(L_{11}) = \phi_{\text{KM}}^{\text{cyc},(2)}, \quad \kappa_2(L_{22}) = \phi_{\text{KM}}^{\text{cyc},(1)}.$$

Its $(2, 1)$ -companion ω_{21} is determined by

$$\begin{aligned}\omega_{21}(L_{11}) &:= -\frac{1}{8\pi^2} \lambda_1 \bar{Z}_2 \wedge Z_1 \wedge Z_2, \\ \omega_{21}(L_{12}) &:= \frac{1}{16\pi^2} (\lambda_1 \bar{Z}_1 - \lambda_2 \bar{Z}_2) \wedge Z_1 \wedge Z_2, \\ \omega_{21}(L_{22}) &:= \frac{1}{8\pi^2} \lambda_2 \bar{Z}_1 \wedge Z_1 \wedge Z_2.\end{aligned}\tag{6.64}$$

Thus

$$\phi_2^{(1,3)} = (\kappa_2, \omega_{21}).$$

The (2, 2)-component. First define the holomorphic primitive η_2^{hol} by

$$\begin{aligned}\eta_2^{\text{hol}}(L_{11} \wedge L_{12}) &:= \frac{1}{4\pi} \lambda_1 Z_2, \\ \eta_2^{\text{hol}}(L_{11} \wedge L_{22}) &:= \frac{1}{4\pi} (\lambda_2 Z_2 - \lambda_1 Z_1), \\ \eta_2^{\text{hol}}(L_{12} \wedge L_{22}) &:= -\frac{1}{4\pi} \lambda_2 Z_1.\end{aligned}\tag{6.65}$$

Its anti-real part is

$$\eta_2 := \eta_2^{\text{hol}} - \overline{\eta_2^{\text{hol}}}.\tag{6.66}$$

The holomorphic two-form component ω_2 is determined by

$$\begin{aligned}\omega_2(L_{11} \wedge L_{12}) &:= \frac{1}{32\pi^2} \lambda_1^2 Z_1 \wedge Z_2, \\ \omega_2(L_{11} \wedge L_{22}) &:= \frac{1}{16\pi^2} \lambda_1 \lambda_2 Z_1 \wedge Z_2, \\ \omega_2(L_{12} \wedge L_{22}) &:= \frac{1}{32\pi^2} \lambda_2^2 Z_1 \wedge Z_2.\end{aligned}\tag{6.67}$$

The middle value is the product of the two rank-one holomorphic forms:

$$\omega_2(L_{11} \wedge L_{22}) = \phi_{\text{KM}}^{\text{hol},(1)} \wedge \phi_{\text{KM}}^{\text{hol},(2)}.\tag{6.68}$$

We then put

$$\phi_2^{(2,2)} = (\eta_2, \omega_2).$$

The (3, 1)-component. Finally, define η_3 by

$$\eta_3(L_{11} \wedge L_{12} \wedge L_{22}) := \frac{1}{16\pi} (\lambda_1 \bar{\lambda}_2 - \bar{\lambda}_1 \lambda_2).\tag{6.69}$$

The right-hand side is anti-invariant under complex conjugation, as required.

Proposition 6.14 (The joint cocycle relations)

With the differential convention fixed above, the four cochains satisfy

$$\begin{aligned}d_D \phi_2^{(0,4)} &= 0, & d_D \phi_2^{(1,3)} &= d_{\text{Sp}} \phi_2^{(0,4)}, \\ d_D \phi_2^{(2,2)} &= d_{\text{Sp}} \phi_2^{(1,3)}, & d_D \phi_2^{(3,1)} &= d_{\text{Sp}} \phi_2^{(2,2)}, \\ d_{\text{Sp}} \phi_2^{(3,1)} &= 0.\end{aligned}\tag{6.70}$$

Consequently,

$$d_{\text{Tot}} \Phi = 0,$$

and Φ defines a total cohomology class of degree four.

6.6 The cohomological theta realization

Let

$$C_{\text{coh}}^a(X_r^{\text{tor}}, \omega_r^{1+n/2}) := A^{0,a}(X_r^{\text{tor}}, \omega_r^{1+n/2})$$

be the Dolbeault complex computing coherent cohomology on the Siegel side, and let

$$C_{\text{DB}}^b(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(r))$$

be the Deligne–Beilinson complex on the orthogonal side. We suppress the finite level from the notation. Fix

$$\varphi_f \in \mathcal{S}(V^r(\mathbb{A}_f)),$$

and let

$$I_\infty : M \longrightarrow \mathcal{S}(V^r(\mathbb{R}))$$

be the Fock–Schrödinger intertwining map. For $P \in M$, put

$$\theta_{\varphi_f, P}(g, h) := \sum_{x \in V^r(\mathbb{Q})} [\omega(g, h)(I_\infty(P) \otimes \varphi_f)](x). \quad (6.71)$$

For

$$\phi \in C_{\text{joint}}^{a,b}(M(r)), \quad X \in \wedge^a \mathfrak{p}_r^-,$$

write

$$\phi(X) = \sum_{\nu} P_\nu \otimes \eta_\nu,$$

where $P_\nu \in M$ and η_ν is an orthogonal Deligne–Beilinson cochain.

Definition 6.15. Define

$$\Theta_{\varphi_f}(\phi)(g, h; X) := \sum_{\nu} \theta_{\varphi_f, P_\nu}(g, h) \eta_\nu. \quad (6.72)$$

Thus theta summation acts on the Fock coefficient and preserves the Deligne–Beilinson form or current factor. In bidegree (a, b) , this gives

$$\Theta_{\varphi_f}^{a,b} : C_{\text{joint}}^{a,b}(M(r)) \longrightarrow C_{\text{coh}}^a(X_r^{\text{tor}}, \omega_r^{1+n/2}) \otimes C_{\text{DB}}^b(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(r)). \quad (6.73)$$

Proposition 6.16

The theta realization is compatible with the two differentials:

$$(\bar{\partial}_{X_r} \otimes 1) \Theta_{\varphi_f}^{a,b}(\phi) = \Theta_{\varphi_f}^{a+1,b}(\text{d}_{\text{Sp}} \phi), \quad (6.74)$$

$$(1 \otimes \text{d}_D) \Theta_{\varphi_f}^{a,b}(\phi) = \Theta_{\varphi_f}^{a,b+1}(\text{d}_D \phi). \quad (6.75)$$

Hence it defines a morphism of total complexes

$$\Theta_{\varphi_f} : C_{\text{Tot}}^\bullet(M(r)) \longrightarrow \text{Tot}\left(C_{\text{coh}}^\bullet(X_r^{\text{tor}}, \omega_r^{1+n/2}) \otimes C_{\text{DB}}^\bullet(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(r))\right). \quad (6.76)$$

Proof. This follows from the intertwining property of the Weil representation and may be checked termwise in (6.71). $\square$

If

$$\Phi = \sum_{a+b=m} \phi^{(a,b)}$$

is a total cocycle, then $\Theta_{\varphi_f}(\Phi)$ is a total cocycle. We denote its (a, b) -Künneth component by

$$[\Theta_{\varphi_f}(\Phi)]^{(a,b)} \in H^a(X_r^{\text{tor}}, \omega_r^{1+n/2}) \otimes H_{\text{DB}}^b(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(r)). \quad (6.77)$$

Remark 6.17

A single component $\phi^{(a,b)}$ need not be closed by itself. Thus $\Theta_{\varphi_f}(\phi^{(a,b)})$ is naturally a cochain, whereas the canonical cohomology class is

$$[\Theta_{\varphi_f}(\Phi)]^{(a,b)}.$$

For $r = 2$, write

$$\Phi = \phi^{(0,4)} + \phi^{(1,3)} + \phi^{(2,2)} + \phi^{(3,1)}.$$

The coherent–Deligne–Beilinson theta class used below is

$$K^{(13)} := [\Theta_{\varphi_f}(\Phi)]^{(1,3)} \in H^1(X_2^{\text{tor}}, \omega_2^{1+n/2}) \otimes H_{\text{DB}}^3(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2)). \quad (6.78)$$

Definition 6.18. For

$$\eta \in H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)),$$

its *Deligne–Beilinson theta lift* is defined by Serre duality:

$$\Theta_{\Phi}^{\text{DB}}(\eta) := \langle K^{(13)}, \eta \rangle_{\text{Serre}} \in H_{\text{DB}}^3(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2)). \quad (6.79)$$

Remark 6.19

Thus $K^{(13)}$ is the universal coherent–Deligne–Beilinson kernel, while $\Theta_{\Phi}^{\text{DB}}(\eta)$ is the theta lift of the particular coherent cohomology class η .

6.7 The motivic Thom property

Recall the staircase of joint Kudla–Millson–Deligne–Beilinson cochains:

$$\begin{array}{c}
 \phi^{(04)} \downarrow d_{\text{Sp}} \\
 \phi^{(13)} \xrightarrow{d_D} 0 \\
 \downarrow d_{\text{Sp}} \\
 \phi^{(22)} \xrightarrow{d_D} 0 \\
 \downarrow d_{\text{Sp}} \\
 \phi^{(31)} \xrightarrow{d_D} 0 \\
 \downarrow d_{\text{Sp}} \\
 0
 \end{array} . \quad (6.80)$$

Applying Θ_{φ_f} gives the corresponding theta staircase:

$$\begin{array}{ccc}
 & & \Theta_{\varphi_f}(\phi^{(04)}) \\
 & & \downarrow d_{Sp} \\
 & \Theta_{\varphi_f}(\phi^{(13)}) & \xrightarrow{d_D} 0 \\
 & \downarrow d_{Sp} \\
 & \Theta_{\varphi_f}(\phi^{(22)}) & \xrightarrow{d_D} 0 \\
 & \downarrow d_{Sp} \\
 \Theta_{\varphi_f}(\phi^{(31)}) & \xrightarrow{d_D} 0 \\
 \downarrow d_{Sp} \\
 0
 \end{array} \tag{6.81}$$

Notation 6.20. • Write Φ for the total class:

$$\Phi = \phi^{(04)} + \phi^{(13)} + \phi^{(22)} + \phi^{(31)}. \tag{6.82}$$

- The Künneth decomposition gives

$$[\Theta_{\varphi_f}(\Phi)] = K^{(04)} + K^{(13)} + K^{(22)} + K^{(31)}, \tag{6.83}$$

where

$$K^{(ij)} \in H^i(X_2^{\text{tor}}, \omega_2^{1+n/2}) \otimes H_{\text{DB}}^j(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2)) \tag{6.84}$$

is the corresponding Künneth component. We write

$$K = K^{(04)} + K^{(13)} + K^{(22)} + K^{(31)}. \tag{6.85}$$

Proposition 6.21

For every $\eta \in H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D))$, the following identity holds in $H_D^3(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2))$:

$$\langle \Theta_{\varphi_f}(\phi^{(13)}), \eta \rangle = \langle K^{(13)}, \eta \rangle, \tag{6.86}$$

where

$$\langle \cdot, \cdot \rangle : H^2(X_2^{\text{tor}}, \omega_2^{2-n/2}(-D)) \times H^1(X_2^{\text{tor}}, \omega_2^{1+n/2}) \longrightarrow \mathbb{C} \tag{6.87}$$

is the Serre duality pairing.

Proof. Since $\Theta_{\varphi_f}(\Phi)$ and K represent the same cohomology class in the total complex, we may write

$$\Theta_{\varphi_f}(\Phi) - K = dH, \tag{6.88}$$

where d is the differential of the total complex. In bidegree $(1, 3)$, this gives

$$\Theta_{\varphi_f}(\phi^{(13)}) - K^{(13)} = \pm d_{Sp}u \pm d_Dv. \tag{6.89}$$

For the closed form η , one computes

$$\begin{aligned}
 & \langle \Theta_{\varphi_f}(\phi^{(13)}) - K^{(13)}, \eta \rangle \\
 &= \text{Tr} \left((\Theta_{\varphi_f}(\phi^{(13)}) - K^{(13)}) \cup \eta \right)
 \end{aligned} \tag{6.90}$$

$$\begin{aligned}
&= \pm \operatorname{Tr}(\mathrm{d}_{\mathrm{Sp}} u \cup \eta) \pm \operatorname{Tr}(\mathrm{d}_D v \cup \eta) \\
&= \pm \operatorname{Tr}(u \cup \mathrm{d}_{\mathrm{Sp}} \eta) \pm \operatorname{Tr}(\mathrm{d}_D(\eta \cup v)) \quad (\text{Stokes' theorem}) \\
&= \pm \operatorname{Tr}(\mathrm{d}_D(\eta \cup v)) \quad (\eta \text{ is closed}) \\
&= \pm \operatorname{Tr}([0]) \\
&= 0.
\end{aligned} \tag{6.91}$$

Therefore

$$\langle \Theta_{\varphi_f}(\phi^{(13)}), \eta \rangle = \langle K^{(13)}, \eta \rangle. \tag{6.92}$$

□

Theorem 6.22 (Motivic Thom property)

The Beilinson regulator of $\Theta^{\mathrm{mot},1}$ is given by $K^{(13)}$:

$$r_D(\Theta^{\mathrm{mot},1}) = \Theta^{\mathrm{DB},1} = K^{(13)}. \tag{6.93}$$

Proof. Both $\Theta^{\mathrm{DB},1}$ and $K^{(13)}$ lie in $H^1(X_2^{\mathrm{tor}}, \omega_2^{1+n/2}) \otimes H_{\mathrm{DB}}^3(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{R}(2))$. By the perfectness of Serre duality, it suffices to show that, for every $\eta \in H^2(X_2^{\mathrm{tor}}, \omega_2^{2-n/2}(-D))$,

$$\langle \Theta^{\mathrm{DB},1}, \eta \rangle = \langle K^{(13)}, \eta \rangle = \langle \Theta_{\varphi_f}(\phi^{(13)}), \eta \rangle. \tag{6.94}$$

By Proposition 4.1 and Remark 4.3, choose a representative

$$\eta = \sum_{m \geq 0} f_m q_{22}^{-m}, \tag{6.95}$$

where

$$f_m \in H^0(\hat{X}_1^{\min}, R^1 \pi_* \kappa^m \otimes \omega_1^{1-n/2}). \tag{6.96}$$

As in Section 5.3, choose a lift

$$\eta' = \sum_{m \geq 0} f'_m q_{22}^{-m} \in H_{X_1^{\mathrm{tor}}}^2(X_2^{\mathrm{tor}}, \omega_2^{2-n/2}(-D)) \tag{6.97}$$

of η , where $f'_m \in H^0(\hat{X}_1^{\min}, R^1 \pi_* \kappa^{-m} \otimes \omega_1^{1-n/2}(-D))$ and the restriction of f'_m to $\hat{X}_1^{\min}$ is f_m .

By the construction of $\phi^{(13)}$, we have

$$\Theta_{\varphi_f}(\phi^{(13)}) = (\Theta_{\varphi_f}(\kappa_2), \Theta_{\varphi_f}(\omega_{21})). \tag{6.98}$$

By Remark 6.4, the class

$$\Theta^{\mathrm{DB},1} = (S, T) \tag{6.99}$$

is determined by S . It therefore suffices to show

$$\langle \Theta_{\varphi_f}(\kappa_2), \eta \rangle = \langle S, \eta \rangle. \tag{6.100}$$

We first compute $\langle \Theta_{\varphi_f}(\kappa_2), \eta \rangle$. In the classical realization of the adelic construction, the Taylor expansion of $\Theta_{\varphi_f}(\kappa_2)$ around $q_{22} = 0$ is

$$\begin{aligned}
\Theta_{\varphi_f}(\kappa_2) &= \sum_{m \geq 0} \Theta_{\varphi_f}^m(\kappa_2) q_{22}^m \\
&= \sum_{m \geq 0} \sum_{\substack{\lambda_2 \in \Lambda \\ (\lambda_2, \lambda_2) = m}} \Theta_{\varphi_f}^{\lambda_2}(\kappa_2) q_{22}^m.
\end{aligned}$$

Thus

$$\begin{aligned}
 \langle \Theta_{\varphi_f}(\kappa_2), \eta \rangle &= \langle \Theta_{\varphi_f}(\kappa_2), \eta' \rangle \\
 &= \sum_{m \geq 0} \sum_{\substack{\lambda_2 \in \Lambda \\ (\lambda_2, \lambda_2) = m}} \int_{X_1^{\text{tor}}} \Theta_{\varphi_f}^{\lambda_2}(\kappa_2) \wedge f'_m \\
 &= \sum_{m \geq 0} \sum_{\substack{\lambda_2 \in \Lambda \\ (\lambda_2, \lambda_2) = m}} \int_{X_1^{\text{tor}}} \Theta_{\varphi_f}^{\lambda_2}(\phi_{\text{KM}}^{\text{cyc}, (2)}) f'_m \, d\tau_{11} \, d\tau_{12},
 \end{aligned}$$

Here the last identity follows from the construction of κ_2 and the fact that $f'_m \in H^0(X_1^{\text{min}}, R^1 \pi_* \kappa^{-m} \otimes \omega_1^{1-n/2}(-D))$. Denote the resulting current $\langle \Theta_{\varphi_f}(\kappa_2), \eta \rangle$ on $\text{Sh}_{\text{SO}(V)}$ by S . Thus

$$S := \left[\sum_{m \geq 0} \sum_{\substack{\lambda_2 \in \Lambda \\ (\lambda_2, \lambda_2) = m}} \int_{X_1^{\text{tor}}} \Theta_{\varphi_f}^{\lambda_2}(\phi_{\text{KM}}^{\text{cyc}, (2)}) f'_m \, d\tau_{11} \, d\tau_{12} \right]. \quad (6.101)$$

It follows from the construction of $\Theta^{\text{Ger}, 1}$ that

$$R := \langle \Theta^{\text{DB}, 1}, \eta \rangle = \sum_{m \geq 0} \sum_{\substack{\lambda_2 \in \Lambda \\ (\lambda_2, \lambda_2) = m}} \left(\int_{X_1^{\text{tor}}}^* f_m \Theta^{\lambda_2} \, d\tau_{11} \, d\tau_{12} \right) \delta_{Z(\lambda_2)}. \quad (6.102)$$

It remains to show that

$$S = R, \quad (6.103)$$

or equivalently that, for every $\omega \in \Omega^{n-1, n-1}(\text{Sh}_{\text{SO}(V)})$, one has

$$\langle S, \omega \rangle = \langle R, \omega \rangle. \quad (6.104)$$

Let

$$\iota_{Z(\lambda_2)} : Z(\lambda_2) \longrightarrow \text{Sh}_{\text{SO}(V)} \quad (6.105)$$

be the closed embedding of the special divisor $Z(\lambda_2)$. The (m, λ_2) -summand of $\langle R, \omega \rangle$ is

$$\begin{aligned}
 \langle R, \omega \rangle_{m, \lambda_2} &= \int_{Z(\lambda_2)} \left(\int_{X_1^{\text{tor}}}^* f_m \theta^{\lambda_2} \, d\tau_{11} \, d\tau_{12} \right) \wedge \iota_{Z(\lambda_2)}^* \omega \, dx \\
 &= \int_{\Gamma_{\lambda_2} \setminus \mathbb{D}_{\lambda_2}^+} \int_{X_1^{\text{tor}}}^* f_m \theta^{\lambda_2} \, d\tau_{11} \, d\tau_{12} \wedge \iota_{Z(\lambda_2)}^* \omega \, dx \quad (6.106)
 \end{aligned}$$

$$= \int_{\Gamma_{\lambda_2} \setminus \mathbb{D}_{\lambda_2}^+} \iota_{Z(\lambda_2)}^* \left(\int_{X_1^{\text{tor}}}^* f_m \Theta^{\lambda_2} \, d\tau_{11} \, d\tau_{12} \wedge \omega \right) dx \quad (6.107)$$

$$= \int_{\Gamma_{\lambda_2} \setminus \mathbb{D}^+} \left(\int_{X_1^{\text{tor}}}^* f_m \Theta^{\lambda_2} \, d\tau_{11} \, d\tau_{12} \wedge \omega \right) \wedge \phi_{\text{KM}}^{\text{cyc}, \lambda_2} \, dx \quad (\text{Thom property}) \quad (6.108)$$

$$= \int_{\Gamma \setminus \mathbb{D}^+} \left(\int_{X_1^{\text{tor}}}^* f_m \Theta^{\lambda_2} \, d\tau_{11} \, d\tau_{12} \wedge \omega \right) \wedge \left(\sum_{\Gamma_{\lambda_2} \setminus \Gamma} \gamma^* \phi_{\text{KM}}^{\text{cyc}, \lambda_2} \right) dx \quad (6.109)$$

$$= \int_{\Gamma \setminus \mathbb{D}^+} \left(\int_{X_1^{\text{tor}}}^* f_m \Theta^{\lambda_2}(\phi_{\text{KM}}^{\text{cyc}, (2)}) \, d\tau_{11} \, d\tau_{12} \right) \wedge \omega \, dx \quad (6.110)$$

$$= \int_{\Gamma \setminus \mathbb{D}^+} \left(\int_{X_1^{\text{tor}}}^* f'_m \Theta^{\lambda_2}(\phi_{\text{KM}}^{\text{cyc}, (2)}) \, d\tau_{11} \, d\tau_{12} \right) \wedge \omega \, dx \quad (6.111)$$

$$= \int_{\Gamma \setminus \mathbb{D}^+} \left(\int_{X_1^{\text{tor}}}^* f'_m \Theta^{\lambda_2}(\phi_{\text{KM}}^{\text{cyc}, (2)}) \, d\tau_{11} \, d\tau_{12} \right) \wedge \omega \, dx \quad (6.112)$$

$$= \langle S, \omega \rangle_{m, \lambda_2}. \quad (6.113)$$

Thus $S = R$ as currents, so

$$\Theta^{\text{DB},1} = K^{(13)}. \quad (6.114)$$

□

Remark 6.23

The formula $r_D(\Theta^{\text{mot},1}) = K^{(13)}$ can be viewed as a motivic analogue of the identity

$$\text{cl}(\Theta^a) = \Theta^{\text{KM}}, \quad (6.115)$$

where cl is the cycle class map and Θ^{KM} is the Kudla–Millson theta series defined in [KM90]. Kudla and Millson’s proof relies on the Thom property of their classes, which motivates the name *motivic Thom property*.

7 The regulator pairing and the joint Kudla–Millson classes

7.1 Motivation

First recall the following proposition.

Proposition 7.1 ([BGCLRJ24a, Proposition 2.27])

Let

$$\eta \in A_{\text{rd}}^{2n-2}(\text{Sh}_{\text{SO}(V)})$$

be a smooth closed rapidly decreasing differential form whose Hodge-type components lie in

$$\{(a, b) : a, b > 0\}.$$

Then the assignment

$$(S, T) \mapsto S(\eta)$$

induces a map

$$\langle -, \eta \rangle : H_{\text{DB}}^3(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2)) \longrightarrow \mathbb{C}.$$

To relate $r_D(\Theta^{\text{mot},1})$ to L -values, we need a form η that pairs with it as in the proposition. This section is devoted to the construction of such a form.

7.2 Representation-theoretic preliminaries

7.2.1 The real theta correspondence for $(\text{Sp}_4, \text{SO}(2, 2)^+)$

Recall the exceptional isomorphism of real Lie groups

$$\text{SO}(2, 2)(\mathbb{R})^+ \simeq \frac{\text{SL}_2(\mathbb{R}) \times \text{SL}_2(\mathbb{R})}{\{\pm 1\}}. \quad (7.1)$$

The L -packet of discrete series representations Π_∞ of $\text{SO}(2, 2)(\mathbb{R})^+$ satisfying

$$H^i(\mathfrak{g}, K_\infty; \Pi_\infty) = \begin{cases} 0 & i \neq 2 \\ \mathbb{C} & i = 2 \end{cases} \quad (7.2)$$

consists of the following four representations, with the indicated Hodge types:

representation	Hodge type
$D_2^+ \boxtimes D_2^+$	(2, 0)
$D_2^+ \boxtimes D_2^-$	(1, 1)
$D_2^- \boxtimes D_2^+$	(1, 1)
$D_2^- \boxtimes D_2^-$	(0, 2).

(7.3)

Notation 7.2. Write $\Pi_\infty = D_2^- \boxtimes D_2^+$ and $\Pi'_\infty = D_2^+ \boxtimes D_2^-$, so that Π'_∞ is the contragredient of Π_∞ .

Proposition 7.3

The representation Π_∞ (resp. Π'_∞) is the small theta lift of a limit of discrete series π_∞ (resp. π'_∞) of $\mathrm{Sp}_4(\mathbb{R})$ with Blattner parameter $(0, -2)$ (resp. $(2, 0)$).

Proof. Let $D_{BL}(\Lambda_1, \Lambda_2)$ be a limit of discrete series of $\mathrm{Sp}_4(\mathbb{R})$ with Blattner parameter (Λ_1, Λ_2) . Then it is the small theta lift of $D_{k_1}^- \boxtimes D_{k_2}^+$ (resp. $D_{k_1}^+ \boxtimes D_{k_2}^-$) if and only if

$$(\Lambda_1, \Lambda_2) = \left(\frac{k_1 - k_2}{2}, -\frac{k_1 + k_2}{2} \right) \quad (\text{resp. } (\Lambda_1, \Lambda_2) = \left(\frac{k_1 + k_2}{2}, \frac{k_1 - k_2}{2} \right)) \quad (7.4)$$

For $k_1 = k_2 = 2$, this gives

$$(\Lambda_1, \Lambda_2) = (0, -2) \quad (\text{resp. } (\Lambda_1, \Lambda_2) = (2, 0)). \quad (7.5)$$

□

Notation 7.4. Let

$$\xi_{a,b} := \mathrm{Sym}^{a-b} \mathbb{E}^\vee \otimes (\det \mathbb{E}^\vee)^b, \quad a \geq b. \quad (7.6)$$

be the indicated algebraic representation of K_∞ , where $\mathbb{E}$ is the standard two-dimensional representation.

Proposition 7.5

We have

$$H^2(\mathfrak{q}, K_\infty; \pi_\infty \otimes \xi_{1,1}) \cong \mathbb{C}, \quad (7.7)$$

$$H^1(\mathfrak{q}, K_\infty; \pi'_\infty \otimes \xi_{2,2}) \cong \mathbb{C}. \quad (7.8)$$

Proof. The definition of $(\mathfrak{q}, K_\infty)$ -cohomology and the calculation of K_∞ -types give

$$H^2(\mathfrak{q}, K_\infty; \pi_\infty \otimes \xi_{1,1}) \cong \mathrm{Hom}_{K_\infty}(\wedge^2 p^-, \pi_\infty \otimes \xi_{1,1}) \cong \mathbb{C},$$

$$H^1(\mathfrak{q}, K_\infty; \pi'_\infty \otimes \xi_{2,2}) \cong \mathrm{Hom}_{K_\infty}(p^-, \pi'_\infty \otimes \xi_{2,2}) \cong \mathbb{C}.$$

□

7.2.2 The global theta correspondence for $(\mathrm{Sp}_4, \mathrm{SO}(V))$

Proposition 7.6

There exists a cuspidal automorphic representation

$$\pi = \pi_f \otimes \pi_\infty$$

of $\mathrm{Sp}_4(\mathbb{A})$ such that π_∞ is a limit of discrete series with Blattner parameter $(0, -2)$, and whose theta lift

$$\Pi = \theta(\pi)$$

is a cuspidal automorphic representation of $\mathrm{SO}(V)(\mathbb{A})$ with archimedean component

$$\Pi_\infty = D_2^- \boxtimes D_2^+.$$

Put

$$\pi' := \pi^\vee, \quad \Pi' := \theta(\pi').$$

Then π'_∞ is a limit of discrete series with Blattner parameter $(2, 0)$, and

$$\Pi'_\infty = D_2^+ \boxtimes D_2^-.$$

Moreover, Π' is cuspidal.

Proof. It suffices to prove the existence of π and Π ; the assertions for $\pi' = \pi^\vee$ and $\Pi' = \theta(\pi')$ follow immediately from contragredience.

First, [Shi12, Theorem 4.8] gives a cuspidal automorphic representation $\Pi = \Pi_f \otimes \Pi_\infty$ of $\mathrm{SO}(V)(\mathbb{A})$ with $\Pi_\infty = D_2^- \boxtimes D_2^+$. We choose two finite places v_1 and v_2 such that Π_{v_1} is square-integrable, while Π_{v_2} is supercuspidal and has zero small theta lift to SL_2 but nonzero small theta lift to Sp_4 . In other words, Sp_4 is the first occurrence of Π_{v_2} .

Next, by [Tak09, Proposition A.2], for every finite place $v \neq v_2$, the local small theta lift $\theta_v(\Pi_v)$ of Π_v to Sp_4 is nonzero.

By [Tak11, Theorem 1.3(1)], the global representation

$$\pi = \theta(\Pi) = \otimes'_v \theta_v(\Pi_v). \quad (7.9)$$

is nonzero.

Finally, the first-occurrence condition and Rallis's tower property [Ral84, Theorem I.1.1(1)] imply that π is cuspidal. $\square$

We now describe the cohomological realizations of π and Π .

Notation 7.7. The representation $\xi_{1,1}$ (resp. $\xi_{2,2}$) corresponds to the automorphic line bundle $\omega_2(-D)$ (resp. $\omega_2^2(-D)$).

Proposition 7.8

We have

•

$$\pi_f \otimes H^2(\mathfrak{q}, K_\infty; \pi_\infty \otimes \xi_{1,1}) \subset H^2(X_2^{\mathrm{tor}}, \omega_2(-D)), \quad (7.10)$$

$$\pi'_f \otimes H^1(\mathfrak{q}, K_\infty; \pi'_\infty \otimes \xi_{2,2}) \subset H^1(X_2^{\mathrm{tor}}, \omega_2^2(-D)). \quad (7.11)$$

•

$$\Pi_f \otimes H^2(\mathfrak{g}, K_\infty; \Pi_\infty) \subset H_B^2(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{C}), \quad (7.12)$$

$$\Pi'_f \otimes H^2(\mathfrak{g}, K_\infty; \Pi'_\infty) \subset H_B^2(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{C}). \quad (7.13)$$

Proof. This follows directly from Proposition 7.5. $\square$

Definition 7.9. Choose a highest-weight vector v of $\wedge^2 \mathfrak{p}^-$ and a lowest-weight vector v' of $\mathfrak{p}^-$. Let e and e' be the canonical basis vectors of $\xi_{1,1}$ and $\xi_{2,2}$, respectively. By Propositions 7.5 and 7.8, these choices associate to each $\varphi = \varphi_f \otimes \varphi_\infty \in \pi$ (resp. $\varphi' = \varphi'_f \otimes \varphi'_\infty \in \pi'$) a canonical differential form

$$\omega_\varphi := \varphi_f \otimes \omega_{\varphi_\infty} \in H^2(X_2^{\mathrm{tor}}, \omega_2(-D)) \quad (7.14)$$

$$\text{(resp. } \omega_{\varphi'} := \varphi'_f \otimes \omega_{\varphi'_\infty} \in H^1(X_2^{\mathrm{tor}}, \omega_2^2(-D)) \text{)}, \quad (7.15)$$

such that

$$\omega_{\varphi_\infty}(v) = \varphi_\infty \otimes e \quad (7.16)$$

$$(\text{resp. } \omega_{\varphi'_\infty}(v') = \varphi'_\infty \otimes e'). \quad (7.17)$$

Definition 7.10 (Motivic and Deligne–Beilinson theta lifts). The motivic map is

$$H^2(X_2^{\text{tor}}, \xi_{1,1}(-D)) \xrightarrow{\Theta^{\text{mot},1}} \text{CH}^2(\text{Sh}_{\text{SO}(2,2)}, 1) \xrightarrow{r_D} H_{\text{DB}}^3(\text{Sh}_{\text{SO}(2,2)}, \mathbb{R}(2)). \quad (7.18)$$

For $g \in \pi$, applying the motivic map to ω_g defines the *motivic theta lift*

$$\Theta^{\text{mot}}(\omega_g) := \langle \omega_g, \Theta^{\text{mot},1} \rangle, \quad (7.19)$$

and the *Deligne–Beilinson theta lift*

$$\Theta^{\text{DB}}(\omega_g) := \langle \omega_g, r_D(\Theta^{\text{mot},1}) \rangle. \quad (7.20)$$

Remark 7.11

The goal of these notes is to prove that

$$\Theta^{\text{mot}}(\omega_g) \neq 0 \quad (7.21)$$

by showing that

$$\Theta^{\text{DB}}(\omega_g) \neq 0. \quad (7.22)$$

Notation 7.12. We also write $\Theta_{\varphi_f}^{\text{mot}}(\omega_g)$ and $\Theta_{\varphi_f}^{\text{DB}}(\omega_g)$ to indicate the dependence on the finite Schwartz function $\varphi_f \in S(V(\mathbb{A}_f)^{\otimes 2})$.

Notation 7.13. Write $\Pi = \theta(\pi)$ and $\Pi' = \theta(\pi')$ for the global theta lifts of π and π' , respectively. Both are cuspidal automorphic representations of $\text{SO}(V)$.

Proposition 7.14

Let $M(\Pi_f)$ be the motive associated with Π , and let $M_B(\Pi_f)$ and $M_{\text{dR}}(\Pi_f)$ denote its Betti and de Rham realizations, respectively. Then

$$\dim M_B(\Pi_f)^+ = 2, \quad \dim F^0 M_{\text{dR}}(\Pi_f) = 1, \quad \dim(F^0 M_{\text{dR}}(\Pi_f) \setminus M_B^+(\Pi_f)) = 1. \quad (7.23)$$

Proof. This follows directly from the Hodge type of Π_∞ . $\square$

Remark 7.15

The dimension formula $\dim(F^0 M_{\text{dR}}(\Pi_f) \setminus M_B^+(\Pi_f)) = 1$ suggests the possibility of a Beilinson-type regulator formula for $M(\Pi_f)$.

7.3 The joint Betti relative Lie algebra complex

We work over $\mathbb{C}$ and suppress Tate twists. Retaining the contragredient coefficient convention, put

$$N_r := M^\vee \otimes \det(\mathfrak{p}_r^-), \quad \det(\mathfrak{p}_r^-) := \wedge^{r(r+1)/2} \mathfrak{p}_r^-. \quad (7.24)$$

Here $M^\vee$ carries the contragredient action. The determinant line carries the action induced by the adjoint action of $\mathfrak{k}_{r,\mathbb{C}}$ on $\mathfrak{p}_r^-$; both $\mathfrak{p}_r^-$ and $\mathfrak{g}_{V,\mathbb{C}}$ act trivially on this line. We continue to use $\wedge^a \mathfrak{p}_r^-$ as the domain of the symplectic cochains.

The contragredient action is defined by

$$(\rho^\vee(X)\ell)(m) := -\ell(\rho_M(X)m), \quad m \in M, \quad \ell \in M^\vee. \quad (7.25)$$

Here ρ_M includes the coefficient character already incorporated in M . To specify the polynomial realization, write $x_{\nu j}$ for the original Fock coordinates and $w_{\nu j}$ for the dual coordinates, with $1 \leq \nu \leq p+2$.

We use the complex-bilinear Fischer pairing

$$\langle x^I, w^J \rangle_F := \delta_{I,J} (4\pi)^{|I|} I!. \quad (7.26)$$

Thus multiplication by x_{vj} transposes to $4\pi\partial_{w_{vj}}$, and $\partial_{x_{vj}}$ transposes to $w_{vj}/(4\pi)$. In particular, before any change of columns,

$$\rho^\vee(L_{jk}) = \frac{1}{8\pi} \sum_{\alpha=1}^p w_{\alpha j} w_{\alpha k} - 2\pi \sum_{\mu=p+1}^{p+2} \frac{\partial^2}{\partial w_{\mu j} \partial w_{\mu k}}. \quad (7.27)$$

For $r = 2$, we use the following column convention on the contragredient coefficient space:

$$w_{v1} = z_{v2}, \quad w_{v2} = -z_{v1}, \quad 1 \leq v \leq p+2. \quad (7.28)$$

The same change is made in the positive and negative coordinates. Put

$$q_{jk} := \sum_{\alpha=1}^p z_{\alpha j} z_{\alpha k}, \quad \lambda_j := z_{p+1,j} + iz_{p+2,j}, \quad \bar{\lambda}_j := z_{p+1,j} - iz_{p+2,j}.$$

Suppressing the determinant-line generator, write $\mathcal{L}_{jk} := \rho_{N_2}(L_{jk})$. Then

$$\mathcal{L}_{11} = \frac{q_{22}}{8\pi} - 8\pi \frac{\partial^2}{\partial \lambda_2 \partial \bar{\lambda}_2}, \quad (7.29)$$

$$\mathcal{L}_{12} = -\frac{q_{12}}{8\pi} + 4\pi \left(\frac{\partial^2}{\partial \lambda_1 \partial \bar{\lambda}_2} + \frac{\partial^2}{\partial \bar{\lambda}_1 \partial \lambda_2} \right), \quad (7.30)$$

$$\mathcal{L}_{22} = \frac{q_{11}}{8\pi} - 8\pi \frac{\partial^2}{\partial \lambda_1 \partial \bar{\lambda}_1}. \quad (7.31)$$

The determinant twist does not change the L_{jk} -action. On the compact generators it contributes

$$\rho_{N_r}(H_{jk}) = \rho^\vee(H_{jk}) - (r+1)\delta_{jk} \text{ id}. \quad (7.32)$$

The symbols z_{vj} in these formulas are coordinates on the contragredient model. The generators L_{jk} and the cochain domain $\wedge^\bullet \mathfrak{p}_r^-$ remain unchanged.

The orthogonal Betti relative Lie algebra complex is

$$C_B^b(N_r) := \left(N_r \otimes \wedge^b((\mathfrak{p}_V^+)^* \oplus (\mathfrak{p}_V^-)^*) \right)^{\mathfrak{t}_{V,\mathbb{C}}}, \quad d_B := \partial_V + \bar{\partial}_V. \quad (7.33)$$

There is no mapping-fiber shift in $C_B^\bullet(N_r)$: its degree b is the orthogonal differential-form degree.

Definition 7.16. The joint Betti relative Lie algebra complex is the double complex

$$C_{\text{joint},B}^{a,b}(N_r) := \text{Hom}_{\mathfrak{t}_{r,\mathbb{C}}} \left(\wedge^a \mathfrak{p}_r^-, C_B^b(N_r) \right). \quad (7.34)$$

Its two differentials are

$$d_{\text{Sp}} : C_{\text{joint},B}^{a,b}(N_r) \longrightarrow C_{\text{joint},B}^{a+1,b}(N_r), \quad d_B : C_{\text{joint},B}^{a,b}(N_r) \longrightarrow C_{\text{joint},B}^{a,b+1}(N_r). \quad (7.35)$$

The associated total complex and its differential are

$$C_{\text{Tot},B}^m(N_r) := \bigoplus_{a+b=m} C_{\text{joint},B}^{a,b}(N_r), \quad d_{\text{Tot},B}|_{C_{\text{joint},B}^{a,b}(N_r)} := d_{\text{Sp}} + (-1)^a d_B. \quad (7.36)$$

We denote the total cohomology by

$$H_{\text{joint},B}^m(N_r) := H^m \left(C_{\text{Tot},B}^\bullet(N_r) \right). \quad (7.37)$$

Remark 7.17

The symplectic and orthogonal actions on N_r commute. Consequently,

$$d_{\text{Sp}} d_B = d_B d_{\text{Sp}},$$

and the two relative Lie algebra constructions may be performed in either order.

7.4 Symplectic closedness in bidegree (2, 2)

For $r = 2$, put

$$\mathfrak{p}_2^- = \text{Span}_{\mathbb{C}}\{L_{11}, L_{12}, L_{22}\}, \quad \Delta_{\text{Sp}} := L_{11} \wedge L_{12} \wedge L_{22}.$$

At this stage, we construct only a cochain

$$\phi_B^{(22)} \in C_{\text{joint}, B}^{2,2}(N_2) \quad \text{such that} \quad d_{\text{Sp}} \phi_B^{(22)} = 0. \quad (7.38)$$

Thus (2, 2) records symplectic cochain degree two and orthogonal cochain degree two. We impose no closedness condition in the orthogonal direction.

Write

$$\begin{aligned} A &:= \phi_B^{(22)}(L_{11} \wedge L_{12}), \\ B &:= \phi_B^{(22)}(L_{11} \wedge L_{22}), \\ C &:= \phi_B^{(22)}(L_{12} \wedge L_{22}). \end{aligned} \quad (7.39)$$

Since $\mathfrak{p}_2^-$ is abelian and $\wedge^3 \mathfrak{p}_2^-$ is generated by Δ_{Sp} , symplectic closedness is equivalent to the single identity

$$(d_{\text{Sp}} \phi_B^{(22)})(\Delta_{\text{Sp}}) = \rho_{N_2}(L_{11})C - \rho_{N_2}(L_{12})B + \rho_{N_2}(L_{22})A = 0. \quad (7.40)$$

Here ρ_{N_2} denotes the action on the contragredient coefficient module N_2 , extended to orthogonal cochains by acting on their coefficients. In particular, it is not the original action on M .

Such a cochain determines a class

$$[\phi_B^{(22)}]_{\text{Sp}} \in H^2(\mathfrak{q}_2, \mathfrak{k}_{2, \mathbb{C}}; C_B^2(N_2)). \quad (7.41)$$

The coefficient space here is $C_B^2(N_2)$, the space of orthogonal two-cochains, rather than its Betti cohomology.

Remark 7.18

We do not claim at this stage that $\phi_B^{(22)}$ is a cocycle in the total complex. Indeed, regarded as a total-degree-four cochain concentrated in bidegree (2, 2), it satisfies

$$d_{\text{Tot}, B} \phi_B^{(22)} = d_B \phi_B^{(22)},$$

which is not required to vanish. No additional components in bidegrees (0, 4), (1, 3), or (3, 1) are included in the present construction.

7.5 An explicit representative and the correction term

We now specialize to $n = r = 2$, so that V has signature (2, 2), and give an explicit cochain satisfying (7.38). We retain the notation N_2 , ρ_{N_2} , d_{Sp} , and Δ_{Sp} from the preceding subsection, as well as the notation A, B, C in (7.39).

7.5.1 The contragredient Fock realization

Let $\mathcal{P}_2$ be the untwisted polynomial Fock model. We use the Serre-dual coefficient module in the realization

$$\begin{aligned} N_2 &\simeq (\mathcal{P}_2 \otimes \det_{\mathrm{Sp}}^{-2})^\vee \otimes \bigwedge^3 \mathfrak{p}_2^- \\ &\simeq \mathcal{P}_2^\vee \otimes \det_{\mathrm{Sp}}^{-1}. \end{aligned} \quad (7.42)$$

Throughout this subsection, the tensor factor in the external character $\det_{\mathrm{Sp}}^{-1}$ is suppressed.

We fix the following coordinates on the contragredient model. If $\widehat{z}_{\gamma j}$ are the direct dual Fock coordinates, put

$$z_{\gamma 1} := \widehat{z}_{\gamma 2}, \quad z_{\gamma 2} := -\widehat{z}_{\gamma 1}, \quad 1 \leq \gamma \leq 4. \quad (7.43)$$

The notation below refers to these dual coordinates, rather than the original coordinates on M . For $j = 1, 2$, set

$$\begin{aligned} \lambda_j &:= z_{3j} + iz_{4j}, & \bar{\lambda}_j &:= z_{3j} - iz_{4j}, \\ Z_j &:= \sum_{\alpha=1}^2 z_{\alpha j} d\tau_\alpha, & \bar{Z}_j &:= \sum_{\alpha=1}^2 z_{\alpha j} d\bar{\tau}_\alpha. \end{aligned} \quad (7.44)$$

Write

$$Q_{ij} := \sum_{\alpha=1}^2 z_{\alpha i} z_{\alpha j}, \quad U_{ij} := Z_i \wedge \bar{Z}_j. \quad (7.45)$$

In these coordinates, the symplectic operators on N_2 are

$$\begin{aligned} \rho_{N_2}(L_{11}) &= \frac{Q_{22}}{8\pi} - 8\pi \frac{\partial^2}{\partial \lambda_2 \partial \bar{\lambda}_2}, \\ \rho_{N_2}(L_{12}) &= -\frac{Q_{12}}{8\pi} + 4\pi \left(\frac{\partial^2}{\partial \lambda_1 \partial \bar{\lambda}_2} + \frac{\partial^2}{\partial \bar{\lambda}_1 \partial \lambda_2} \right), \\ \rho_{N_2}(L_{22}) &= \frac{Q_{11}}{8\pi} - 8\pi \frac{\partial^2}{\partial \lambda_1 \partial \bar{\lambda}_1}. \end{aligned} \quad (7.46)$$

Thus the differential remains exactly the one in (7.40); the contragredient convention is encoded in ρ_{N_2} .

7.5.2 The correction term

Put

$$\omega := d\tau_1 \wedge d\bar{\tau}_1 + d\tau_2 \wedge d\bar{\tau}_2. \quad (7.47)$$

For convenience, write

$$U_{ij}^\circ := U_{ij} - \frac{1}{2} Q_{ij} \omega. \quad (7.48)$$

First consider the uncorrected cochain $\phi_{B,0}^{(22)}$, given by

$$\begin{aligned} \phi_{B,0}^{(22)}(L_{11} \wedge L_{12}) &:= -\frac{1}{2\pi} U_{22}, \\ \phi_{B,0}^{(22)}(L_{11} \wedge L_{22}) &:= \frac{1}{2\pi} (U_{12} + U_{21}), \\ \phi_{B,0}^{(22)}(L_{12} \wedge L_{22}) &:= -\frac{1}{2\pi} U_{11}. \end{aligned} \quad (7.49)$$

Its two extreme values are the rank-one cycle-form expressions $\phi_{\mathrm{KM}}^{\mathrm{cyc},(2)}$ and $\phi_{\mathrm{KM}}^{\mathrm{cyc},(1)}$, respectively, interpreted in the present contragredient model.

Define the correction term R_0 by

$$\begin{aligned} R_0(L_{11} \wedge L_{12}) &= \frac{Q_{22}}{4\pi} \omega, \\ R_0(L_{11} \wedge L_{22}) &= -\frac{Q_{12}}{2\pi} \omega, \\ R_0(L_{12} \wedge L_{22}) &= \frac{Q_{11}}{4\pi} \omega. \end{aligned} \tag{7.50}$$

We put

$$\boxed{\phi_B^{(22)} := \phi_{B,0}^{(22)} + R_0.} \tag{7.51}$$

In the notation of (7.39), its three values are

$$\boxed{\begin{aligned} A &= -\frac{1}{2\pi} U_{22}^\circ, \\ B &= \frac{1}{2\pi} (U_{12}^\circ + U_{21}^\circ), \\ C &= -\frac{1}{2\pi} U_{11}^\circ. \end{aligned}} \tag{7.52}$$

In particular,

$$\phi_B^{(22)}(L_{12} \wedge L_{22}) = -\frac{1}{2\pi} \left(Z_1 \wedge \bar{Z}_1 - \frac{1}{2} Q_{11} \omega \right). \tag{7.53}$$

7.5.3 Symplectic closedness

Proposition 7.19 (The symplectic representative)

The cochain $\phi_B^{(22)}$ defined above belongs to $C_{\text{joint},B}^{2,2}(N_2)$ and satisfies

$$d_{\text{Sp}} \phi_B^{(22)} = 0. \tag{7.54}$$

It therefore determines a class

$$[\phi_B^{(22)}]_{\text{Sp}} \in H^2(\mathfrak{q}_2, \mathfrak{k}_{2,\mathbb{C}}; C_B^2(N_2)).$$

Proof. In the dual coordinates fixed above, the non-diagonal compact operators act on the positive variables by

$$\begin{aligned} \rho_{N_2}(H_{12})Z_1 &= 0, & \rho_{N_2}(H_{12})Z_2 &= Z_1, \\ \rho_{N_2}(H_{21})Z_1 &= Z_2, & \rho_{N_2}(H_{21})Z_2 &= 0, \end{aligned}$$

and likewise on $\bar{Z}_1, \bar{Z}_2$. The explicit formulas for A, B, C therefore give

$$\begin{aligned} \rho_{N_2}(H_{12})A &= -B, & \rho_{N_2}(H_{21})A &= 0, \\ \rho_{N_2}(H_{12})B &= -2C, & \rho_{N_2}(H_{21})B &= -2A, \\ \rho_{N_2}(H_{12})C &= 0, & \rho_{N_2}(H_{21})C &= -B. \end{aligned} \tag{7.55}$$

Including the suppressed external character, their respective (H_{11}, H_{22}) -weights are

$$(-3, -1), \quad (-2, -2), \quad (-1, -3).$$

These are the weights of the corresponding basis wedges in (7.39). This proves $\mathfrak{k}_{2,\mathbb{C}}$ -equivariance. The contractions defining U_{ij} , Q_{ij} , and ω also give invariance under the diagonal action of the orthogonal maximal compact subgroup on the Fock and form factors. Hence $\phi_B^{(22)} \in C_{\text{joint},B}^{2,2}(N_2)$.

To check closedness, put

$$D := Q_{11}Q_{22} - Q_{12}^2.$$

Since $n = 2$, direct expansion gives

$$Q_{22}U_{11} - Q_{12}(U_{12} + U_{21}) + Q_{11}U_{22} = D\omega. \quad (7.56)$$

Substituting (7.48) yields

$$Q_{22}U_{11}^\circ - Q_{12}(U_{12}^\circ + U_{21}^\circ) + Q_{11}U_{22}^\circ = 0. \quad (7.57)$$

The values A, B, C are independent of $\lambda_j, \bar{\lambda}_j$, so the derivative terms in (7.46) vanish. Thus

$$\begin{aligned} (\mathrm{d}_{\mathrm{Sp}}\phi_B^{(22)})(\Delta_{\mathrm{Sp}}) &= \rho_{N_2}(L_{11})C - \rho_{N_2}(L_{12})B + \rho_{N_2}(L_{22})A \\ &= \frac{1}{8\pi}(Q_{22}C + Q_{12}B + Q_{11}A) \\ &= -\frac{1}{16\pi^2}(Q_{22}U_{11}^\circ - Q_{12}(U_{12}^\circ + U_{21}^\circ) + Q_{11}U_{22}^\circ) \\ &= 0. \end{aligned} \quad (7.58)$$

□

7.6 The cohomological theta realization

We retain $n = r = 2$ and the contragredient coefficient module

$$N_2 \simeq \mathcal{P}_2^\vee \otimes \det_{\mathrm{Sp}}^{-1}.$$

The external character $\det_{\mathrm{Sp}}^{-1}$ corresponds to the line bundle ω_2 on the Siegel side. Accordingly, let

$$C_{\mathrm{coh}}^a(X_2^{\mathrm{tor}}, \omega_2) := A^{0,a}(X_2^{\mathrm{tor}}, \omega_2)$$

be the Dolbeault complex computing coherent cohomology. On the orthogonal side, we use the complexified de Rham realization of Betti cochains:

$$C_B^b(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{C}) := A^b(\mathrm{Sh}_{\mathrm{SO}(V)}, \mathbb{C}).$$

We suppress the finite level from the notation.

The theta formulas are first defined on the open Shimura varieties. For the compactified formulation below, we assume that the theta cochains under consideration extend to the displayed Dolbeault complex on X_2^{tor} . Tensor products of smooth cochain spaces are understood as completed tensor products.

7.6.1 Theta summation

Fix

$$\varphi_f \in \mathcal{S}(V^2(\mathbb{A}_f)),$$

and let

$$I_\infty^\vee : \mathcal{P}_2^\vee \longrightarrow \mathcal{S}(V^2(\mathbb{R}))$$

be the Fock–Schrödinger intertwining map for the contragredient Weil representation $\omega^\vee$. For $P \in \mathcal{P}_2^\vee$, put

$$\theta_{\varphi_f, P}^B(g, h) := \sum_{x \in V^2(\mathbb{Q})} [\omega^\vee(g, h)(I_\infty^\vee(P) \otimes \varphi_f)](x). \quad (7.59)$$

The finite Schwartz space is equipped with the corresponding contragredient Weil action.

Write e_B for a generator of the external character $\det_{\mathrm{Sp}}^{-1}$. For

$$\phi \in C_{\mathrm{joint}, B}^{a,b}(N_2), \quad X \in \bigwedge^a \mathfrak{p}_2^-,$$

write

$$\phi(X) = \sum_{\nu} (P_{\nu} \otimes e_B) \otimes \eta_{\nu},$$

where $P_{\nu} \in \mathcal{P}_2^{\vee}$ and η_{ν} is an orthogonal exterior-form factor.

Definition 7.20. Define

$$\Theta_{\varphi_f, B}(\phi)(g, h; X) := \sum_{\nu} \theta_{\varphi_f, P_{\nu}}^B(g, h) e_B \otimes \eta_{\nu}. \quad (7.60)$$

The expression is written in the homogeneous frames on the two symmetric domains; equivariance gives its descent to the Shimura varieties. Theta summation acts on the Fock coefficient and preserves the orthogonal form factor. In bidegree (a, b) , it gives

$$\Theta_{\varphi_f, B}^{a, b} : C_{\text{joint}, B}^{a, b}(N_2) \longrightarrow C_{\text{coh}}^a(X_2^{\text{tor}}, \omega_2) \widehat{\otimes} C_B^b(\text{Sh}_{\text{SO}(V)}, \mathbb{C}). \quad (7.61)$$

Proposition 7.21

For each orthogonal degree b , the theta realization is compatible with the symplectic differential:

$$(\bar{\partial}_{X_2} \otimes 1) \Theta_{\varphi_f, B}^{a, b}(\phi) = \Theta_{\varphi_f, B}^{a+1, b}(\text{d}_{\text{Sp}} \phi). \quad (7.62)$$

Consequently, it induces

$$\Theta_{\varphi_f, B, *} : H^a(\mathfrak{q}_2, \mathfrak{k}_{2, \mathbb{C}}; C_B^b(N_2)) \longrightarrow H^a(X_2^{\text{tor}}, \omega_2) \otimes C_B^b(\text{Sh}_{\text{SO}(V)}, \mathbb{C}). \quad (7.63)$$

No orthogonal cohomology is taken in this map.

Proof. The intertwining property identifies the action of $\rho_{N_2}(L_{ij})$ on the source with the corresponding antiholomorphic differential operator on the Siegel side. Applying this termwise in (7.59) gives (7.62). Termwise differentiation is justified by the locally uniform convergence of the Schwartz theta series and its derivatives.

Taking cohomology only in the Siegel direction gives (7.63). Since the coherent cohomology of X_2^{tor} is finite-dimensional, the target can be written with the ordinary tensor product displayed there. $\square$

7.6.2 The (2, 2) theta class

For the cochain $\phi_B^{(22)}$ constructed above,

$$\text{d}_{\text{Sp}} \phi_B^{(22)} = 0.$$

Hence

$$(\bar{\partial}_{X_2} \otimes 1) \Theta_{\varphi_f, B}(\phi_B^{(22)}) = 0.$$

We define

$$\begin{aligned} K_B^{(22)} &:= \left[\Theta_{\varphi_f, B}(\phi_B^{(22)}) \right]_{\text{Sp}} \\ &\in H^2(X_2^{\text{tor}}, \omega_2) \otimes A^{1,1}(\text{Sh}_{\text{SO}(V)}, \mathbb{C}). \end{aligned} \quad (7.64)$$

Here the subscript Sp means that cohomology is taken only in the Siegel direction. The orthogonal factor remains a smooth $(1, 1)$ -form, not a Betti cohomology class.

Remark 7.22

Unlike a component of a general total cocycle, the single cochain $\phi_B^{(22)}$ is already closed in the symplectic direction. Thus (7.64) does not require additional components or a Künneth projection

from total cohomology. We do not assert that

$$(1 \otimes d_B) \Theta_{\varphi_f, B}(\phi_B^{(22)}) = 0.$$

7.6.3 The geometric theta lift via Serre duality

Let D denote the boundary divisor of X_2^{tor} . The canonical-bundle identity

$$\Omega_{X_2^{\text{tor}}}^3 \simeq \omega_2^3(-D)$$

gives the Serre pairing

$$H^2(X_2^{\text{tor}}, \omega_2) \times H^1(X_2^{\text{tor}}, \omega_2^2(-D)) \longrightarrow \mathbb{C}. \quad (7.65)$$

Definition 7.23. For

$$\eta \in H^1(X_2^{\text{tor}}, \omega_2^2(-D)),$$

define its theta lift by

$$\Theta_{\varphi_f, \phi_B}^B(\eta) := \left\langle K_B^{(22)}, \eta \right\rangle_{\text{Serre}} \in A^{1,1}(\text{Sh}_{\text{SO}(V)}, \mathbb{C}). \quad (7.66)$$

If $\tilde{\eta}$ is a Dolbeault representative of η , then this is equivalently

$$\Theta_{\varphi_f, \phi_B}^B(\eta) = \int_{X_2^{\text{tor}}} \Theta_{\varphi_f, B}^{2,2}(\phi_B^{(22)}) \wedge \tilde{\eta}. \quad (7.67)$$

The coefficient bundles are combined using

$$\omega_2 \otimes \omega_2^2(-D) \simeq \Omega_{X_2^{\text{tor}}}^3,$$

and integration is performed only over the Siegel variable. Symplectic closedness ensures that (7.67) depends only on the coherent cohomology class η .

Remark 7.24

Thus $K_B^{(22)}$ is the universal kernel with coherent cohomology on the Siegel side and smooth $(1, 1)$ -forms on the orthogonal side, while $\Theta_{\varphi_f, \phi_B}^B(\eta)$ is its value on a particular coherent cohomology class. The superscript B records the orthogonal cochain realization; it does not assert that the resulting form is closed.

Proposition 7.25

For every $g' \in \pi'$, the geometric theta lift $\Theta_{\varphi_f, \phi_B}^B(\omega_{g'})$ defines a class in $H_B^2(\text{Sh}_{\text{SO}(V)}, \mathbb{C})[\theta(\pi')]$.

Proof. It follows from the construction that

$$\Theta_{\varphi_f, \phi_B}^B(\omega_{g'}) \in \text{Hom}_{K_{V, \infty}}(\wedge^2 \mathfrak{p}_V, \Pi'_{\infty}).$$

The conclusion follows from the identifications

$$H^2(\mathfrak{g}_V, K_{V, \infty}; \Pi'_{\infty}) \cong \text{Hom}_{K_{V, \infty}}(\wedge^2 \mathfrak{p}_V, \Pi'_{\infty})$$

and

$$H_B^2(\text{Sh}_{\text{SO}(V)}, \mathbb{C})[\Pi'] \cong \Pi'_f \otimes H^2(\mathfrak{g}_V, K_{V, \infty}; \Pi'_{\infty}).$$

□

7.6.4 Vanishing of the correction in the geometric theta lift

Let $G = \mathrm{Sp}_4$, let K_2 be its maximal compact subgroup, and write $X_2 = \mathrm{Sh}_G$ at the level under consideration. The theta kernels below use the contragredient Weil representation, with fixed finite Schwartz data. The external coefficient bundles are contracted in the Serre pairing.

Proposition 7.26 (Vanishing of the correction term)

Let π' be the cuspidal automorphic representation of G chosen above. It satisfies

$$(\pi'_\infty)^{K_2} = 0.$$

Let $\omega_{g'}$ be the associated canonical differential form. Then, as an identity of forms on the orthogonal Shimura variety,

$$\int_{X_2} \Theta(R_0) \wedge \omega_{g'} = 0. \quad (7.68)$$

Consequently,

$$\int_{X_2} \Theta(\phi_B^{(22)}) \wedge \omega_{g'} = \int_{X_2} \Theta(\phi_{B,0}^{(22)}) \wedge \omega_{g'}. \quad (7.69)$$

Proof. Let $v_0 = \mathbf{1}$ be the vacuum in the untwisted contragredient Fock model $\mathcal{P}_2^\vee$, and write $\rho^\vee$ for its infinitesimal action. Its K_2 -character is trivial because V has signature $(2, 2)$. Put

$$\mathcal{V}_0 := U(\mathfrak{sp}_{4,\mathbb{C}})v_0.$$

The external character in N_2 has trivial $\mathfrak{p}_2^-$ -action, so (7.46) gives

$$\rho^\vee(L_{11})v_0 = \frac{Q_{22}}{8\pi}, \quad \rho^\vee(L_{12})v_0 = -\frac{Q_{12}}{8\pi}, \quad \rho^\vee(L_{22})v_0 = \frac{Q_{11}}{8\pi}. \quad (7.70)$$

Thus every Schwartz coefficient of R_0 belongs to $\mathcal{V}_0$.

For a fixed orthogonal adelic parameter h , define the scalar pairing

$$\mathcal{B}_h(f, v) := \int_{[G]} f(g) \Theta_v(g, h) \, dg, \quad [G] := G(\mathbb{Q}) \backslash G(\mathbb{A}).$$

This pairing is invariant under the diagonal action of $G(\mathbb{R})$. Since v_0 is K_2 -invariant, averaging over K_2 , with volume one, gives

$$\mathcal{B}_h(f, v_0) = \mathcal{B}_h \left(\int_{K_2} \pi'(k) f \, dk, v_0 \right) = 0. \quad (7.71)$$

For $X \in \mathfrak{sp}_{4,\mathbb{C}}$, equivariance and integration by parts give

$$\mathcal{B}_h(f, \rho^\vee(X)v) = -\mathcal{B}_h(\pi'_\infty(X)f, v). \quad (7.72)$$

Iterating this identity, we obtain

$$\mathcal{B}_h(f, v) = 0 \quad \text{for all } f \in \pi' \text{ and } v \in \mathcal{V}_0.$$

After contracting the external coefficient bundles, the integral $\int_{X_2} \Theta(R_0) \wedge \omega_{g'}$ is a finite linear combination of these scalar periods, multiplied by ω . It therefore vanishes for every h . The comparison identity follows from $\phi_B^{(22)} = \phi_{B,0}^{(22)} + R_0$. $\square$

Remark 7.27

The K_2 -invariance used above concerns the untwisted Weil vacuum, not its tensor product with the external character in N_2 . The correction vanishes under the Siegel theta period. We do not assert that

$$R_0(L_{12} \wedge L_{22}) \wedge U_{22}$$

vanishes or is exact on the orthogonal side. The geometric pairing is still wedge followed by integration; no Fischer contraction is used. As in Remark 7.18, no orthogonal closedness or total cocycle condition is imposed. Nonvanishing of the resulting theta lift is a separate question.

7.7 The Hodge–Deligne doubling pairing

We have now constructed a Deligne–Beilinson cohomology class

$$\Theta^{\text{DB}}(\omega_g) \in H_{\text{DB}}^3(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(2)) \quad (7.73)$$

and a Betti cohomology class

$$\Theta^B(\omega_{g'}) \in H_B^2(\text{Sh}_{\text{SO}(V)}, \mathbb{R}(1)), \quad (7.74)$$

whose Hodge type is $(1, 1)$.

Proposition 7.1 therefore gives the natural pairing

$$\langle \Theta^{\text{DB}}(\omega_g), \Theta^B(\omega_{g'}) \rangle, \quad (7.75)$$

which we call the *Hodge–Deligne doubling pairing*. In the next section, we relate this pairing to non-critical L -values.

8 The connection to the L -value

In this section, we relate the Hodge–Deligne doubling pairing to a non-critical L -value, as predicted by Beilinson’s conjectures.

8.1 The adelic realization of the pairing

Notation 8.1. • Using the canonical identification of $\wedge^4 \mathfrak{p}_V$ with $\mathbb{C}$, set

$$\Phi := \phi_{\text{KM}}^{\text{cyc}, (2)} \wedge \phi_{\text{KM}}^{\text{cyc}, (1)} \in S(V(\mathbb{A})^{\otimes 4}). \quad (8.1)$$

- Let $\Theta_{\Phi}^{\text{Sp}_8}$ be the theta function associated to $\Phi \in S(V(\mathbb{A})^{\otimes 4})$.

Proposition 8.2

The Hodge–Deligne doubling pairing $\langle \Theta^{\text{DB}}(\omega_g), \Theta^B(\omega_{g'}) \rangle$ equals the adelic integral

$$\int_{[\text{SO}(V)]} \int_{[\text{Sp}_4] \times [\text{Sp}_4]} g(x_1) g'(x_2) \Theta_{\Phi}^{\text{Sp}_8}((x_1, x_2); x) \, dx_1 \, dx_2 \, dx. \quad (8.2)$$

Proof. Recall that

$$\Theta^{\text{DB}}(\omega_g) = (\Theta(\kappa_2)(\omega_g), \Theta(\omega_{21})(\omega_g)), \quad (8.3)$$

and the pairing to be computed is

$$\langle \Theta^{\text{DB}}(\omega_g), \Theta^B(\omega_{g'}) \rangle = \langle \Theta(\kappa_2)(\omega_g), \Theta^B(\omega_{g'}) \rangle \quad (8.4)$$

$$= \langle \Theta(\kappa_2)(\omega_g), \Theta(\phi_{B,0}^{(22)})(\omega_{g'}) \rangle \quad (8.5)$$

$$= \langle \Theta(\kappa_2)(\omega_g), \Theta(\phi_B^{(22)})(\omega_{g'}) \rangle \quad (8.6)$$

Let $X_2^{(1)}$ and $X_2^{(2)}$ denote the two copies of the Siegel threefold in this pairing. Then the pairing is the iterated integral

$$\langle \Theta^{\text{DB}}(\omega_g), \Theta^B(\omega_{g'}) \rangle \quad (8.7)$$

$$= \int_{\text{Sh}_{\text{SO}(V)}} \left(\int_{X_2^{(1)}} \omega_g \wedge \Theta(\kappa_2) \, dx_1 \right) \wedge \left(\int_{X_2^{(2)}} \omega_{g'} \wedge \Theta(\phi_B^{(22)}) \, dx_2 \right) \, dx. \quad (8.8)$$

Recall that for $g = g_f \otimes g_\infty \in \pi$ and $g' = g'_f \otimes g'_\infty \in \pi'$, we have

$$\omega_g = g_f \otimes \omega_{g_\infty}, \quad (8.9)$$

$$\omega_{g'} = g'_f \otimes \omega_{g'_\infty}; \quad (8.10)$$

and

$$\omega_{g_\infty}(v) = g_\infty \otimes e \quad (8.11)$$

$$(\text{resp. } \omega_{g'_\infty}(v') = g'_\infty \otimes e'), \quad (8.12)$$

where $v = L_{12} \wedge L_{22}$ is the highest-weight vector of $\wedge^2 \mathfrak{p}^-$ and $v' = L_{11}$ is the lowest-weight vector of $\mathfrak{p}^-$. Thus

$$\int_{X_2^{(1)}} \omega_g \wedge \Theta(\kappa_2) \, dx_1 = \int_{[\text{Sp}_4]} g(x_1) \Theta(\phi_{\text{KM}}^{\text{cyc},(2)})(x_1; x) dx_1; \quad (8.13)$$

$$\int_{X_2^{(2)}} \omega_{g'} \wedge \Theta(\phi_B^{(22)}) \, dx_2 = \int_{[\text{Sp}_4]} g'(x_2) \Theta(\phi_{\text{KM}}^{\text{cyc},(1)})(x_2; x) dx_2. \quad (8.14)$$

Finally, after normalizing the measure on $\text{SO}(V)(\mathbb{A})$, we have

$$\int_{\text{Sh}_{\text{SO}(V)}} \left(\int_{[\text{Sp}_4]} g(x_1) \Theta(\phi_{\text{KM}}^{\text{cyc},(2)})(x_1; x) dx_1 \right) \wedge \left(\int_{[\text{Sp}_4]} g'(x_2) \Theta(\phi_{\text{KM}}^{\text{cyc},(1)})(x_2; x) dx_2 \right) dx \quad (8.15)$$

$$= \int_{[\text{SO}(V)]} \int_{[\text{Sp}_4] \times [\text{Sp}_4]} g(x_1) g'(x_2) \Theta_{\Phi}^{\text{Sp}_8}((x_1, x_2); x) \, dx_1 \, dx_2 \, dx. \quad (8.16)$$

□

Proposition 8.3

The adelic integral

$$\int_{[\text{SO}(V)]} \int_{[\text{Sp}_4] \times [\text{Sp}_4]} g(x_1) g'(x_2) \Theta_{\Phi}^{\text{Sp}_8}((x_1, x_2); x) \, dx_1 \, dx_2 \, dx \quad (8.17)$$

equals the doubling zeta integral

$$Z(s_0, f_{s_0}, g, g') := \int_{[\text{Sp}_4] \times [\text{Sp}_4]} E(\iota(x_1, x_2), s_0, f_{\Phi, s_0}) g(x_1) g'(x_2) \, dx_1 \, dx_2, \quad (8.18)$$

where $s_0 = -1/2$. Here $E(-, s_0, f_{\Phi, s_0})$ is a Siegel Eisenstein series on Sp_8 , with section normalized by

$$f_{\Phi, s_0}(g) = (\omega(g)\Phi)(0) \in \text{Ind}_{P_{\square}(\mathbb{A})}^{\text{Sp}_8(\mathbb{A})}((\chi_V \circ \det) |\det|^{s_0}). \quad (8.19)$$

Proof. Let r be the Witt index of V . For Sp_{2n} and a quadratic space of dimension m , the relevant point is

$$\rho_n = \frac{n+1}{2}, \quad s_0 = \frac{m}{2} - \rho_n. \quad (8.20)$$

In the present doubling construction, this gives

$$s_0 = 2 - \frac{5}{2} = -\frac{1}{2}. \quad (8.21)$$

First suppose that $r = 0$. The theta integral

$$\int_{[\text{SO}(V)]} \Theta_{\Phi}^{\text{Sp}_8}((x_1, x_2); x) dx$$

converges absolutely and equals

$$E(\iota(x_1, x_2), s_0, f_{\Phi, s_0})$$

by the Siegel–Weil formula. The cuspidality of g and g' gives

$$\int_{[\mathrm{Sp}_4] \times [\mathrm{Sp}_4]} \int_{[\mathrm{SO}(V)]} |g(x_1)g'(x_2)\Theta_{\Phi}^{\mathrm{Sp}_8}((x_1, x_2); x)| dx_1 dx_2 dx < \infty. \quad (8.22)$$

Fubini's theorem therefore gives

$$\begin{aligned} & \int_{[\mathrm{SO}(V)]} \int_{[\mathrm{Sp}_4] \times [\mathrm{Sp}_4]} g(x_1)g'(x_2)\Theta_{\Phi}^{\mathrm{Sp}_8}((x_1, x_2); x) dx_1 dx_2 dx \\ &= \int_{[\mathrm{Sp}_4] \times [\mathrm{Sp}_4]} g(x_1)g'(x_2) \int_{[\mathrm{SO}(V)]} \Theta_{\Phi}^{\mathrm{Sp}_8}((x_1, x_2); x) dx_1 dx_2 dx \\ &= \int_{[\mathrm{Sp}_4] \times [\mathrm{Sp}_4]} E(\iota(x_1, x_2), s_0, f_{\Phi, s_0}) g(x_1)g'(x_2) dx_1 dx_2. \end{aligned}$$

Now suppose that $r > 0$. Since $\dim V = 4$, we introduce the regularized theta integral¹

$$\int_{[\mathrm{SO}(V)]}^* \Theta_{\Phi}^{\mathrm{Sp}_8}((x_1, x_2); x) dx \quad (8.23)$$

as in [GQT14, §§ 3.5, 3.7]. The first term identity of the regularized Siegel–Weil formula gives [GQT14, Theorem 7.4]

$$\int_{[\mathrm{SO}(V)]}^* \Theta_{\Phi}^{\mathrm{Sp}_8}((x_1, x_2); x) dx = E(\iota(x_1, x_2), s_0, f_{\Phi, s_0}).$$

The definition of the regularized integral, together with the moderate growth of Eisenstein series and the rapid decay of cusp forms, gives

$$\begin{aligned} & \int_{[\mathrm{SO}(V)]} \int_{[\mathrm{Sp}_4] \times [\mathrm{Sp}_4]} g(x_1)g'(x_2)\Theta_{\Phi}^{\mathrm{Sp}_8}((x_1, x_2); x) dx_1 dx_2 dx \\ &= \int_{[\mathrm{Sp}_4] \times [\mathrm{Sp}_4]} g(x_1)g'(x_2) \int_{[\mathrm{SO}(V)]}^* \Theta_{\Phi}^{\mathrm{Sp}_8}((x_1, x_2); x) dx_1 dx_2 dx \\ &= \int_{[\mathrm{Sp}_4] \times [\mathrm{Sp}_4]} E(\iota(x_1, x_2), s_0, f_{\Phi, s_0}) g(x_1)g'(x_2) dx_1 dx_2. \end{aligned}$$

□

Remark 8.4

Formally, this identity arises from the doubling seesaw

$$\begin{array}{ccc} \mathrm{Sp}(W^+ \oplus W^-) & & \mathrm{SO}(V) \times \mathrm{SO}(V) \\ | & \searrow & | \\ \mathrm{Sp}(W^+) \times \mathrm{Sp}(W^-) & & \mathrm{SO}(V)^{\Delta} \\ \pi \boxtimes \pi^{\vee} & & \mathbf{1} \end{array} \quad (8.24)$$

8.2 The global and local doubling zeta integrals

Notation 8.5. Throughout this subsection, we write

$$m := \dim V,$$

and reserve n for the integer occurring in the symplectic group Sp_{2n} .

¹We define this regularized integral as the residue of the integral in [GQT14, § 3.5], since we are in the first term range; see [GQT14, § 1.7].

Let $\pi = \bigotimes'_v \pi_v$ be an irreducible cuspidal automorphic representation of $\mathrm{Sp}_{2n}(\mathbb{A}_{\mathbb{Q}})$, and choose pure tensors

$$\varphi = \bigotimes_v \varphi_v \in \pi, \quad \tilde{\varphi} = \bigotimes_v \tilde{\varphi}_v \in \pi^\vee.$$

Let $f_s = \bigotimes_v f_{s,v}$ be a factorizable standard section of the normalized Siegel induced representation $I(s, \chi_V)$ on $\mathrm{Sp}_{4n}(\mathbb{A}_{\mathbb{Q}})$.² Recall that the global doubling zeta integral is defined by

$$Z(s, f_s, \varphi, \tilde{\varphi}) := \int_{[\mathrm{Sp}_{2n}] \times [\mathrm{Sp}_{2n}]} E(\iota(g_1, g_2), s, f_s) \varphi(g_1) \tilde{\varphi}(g_2) \, dg_1 \, dg_2. \quad (8.25)$$

Here

$$[\mathrm{Sp}_{2n}] := \mathrm{Sp}_{2n}(\mathbb{Q}) \backslash \mathrm{Sp}_{2n}(\mathbb{A}_{\mathbb{Q}}).$$

We choose compatible Haar measures and normalize the local invariant pairings $\langle -, - \rangle_v$ so that their product is the global Petersson pairing between π and $\pi^\vee$.

Proposition 8.6 (Doubling method [PSR87])

(1) For $\mathrm{Re}(s) \gg 0$, unfolding gives the Euler factorization

$$Z(s, f_s, \varphi, \tilde{\varphi}) = \prod_v Z_v(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v),$$

where

$$Z_v(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v) := \int_{\mathrm{Sp}_{2n}(\mathbb{Q}_v)} f_{s,v}(\iota(g, 1)) \langle \pi_v(g) \varphi_v, \tilde{\varphi}_v \rangle_v \, dg. \quad (8.26)$$

The identity extends meromorphically in s ; see [GQT14, §§ 11.5–11.6].

(2) At every finite place v , the local integral $Z_v(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v)$ converges absolutely for $\mathrm{Re}(s) \gg 0$ and admits continuation to a rational function of q_v^{-s} , where $q_v = p$ for $v = p$; see [Gar15a, Theorem 2.1(1)–(2)].

(3) Suppose that v is finite and that both π_v and χ_v are unramified. Choose spherical data normalized by

$$f_{s,v}(1) = 1, \quad \langle \varphi_v, \tilde{\varphi}_v \rangle_v = 1,$$

and normalize the Haar measure by

$$\mathrm{vol}(\mathrm{Sp}_{2n}(\mathbb{Z}_v)) = 1.$$

Then

$$Z_v(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v) = \frac{L_v(s + \frac{1}{2}, \pi_v \times \chi_v, \mathrm{std})}{d_v(s, \chi_v)}, \quad (8.27)$$

where

$$d_v(s, \chi_v) = L_v\left(s + n + \frac{1}{2}, \chi_v\right) \prod_{j=1}^n \zeta_v(2s + 2j - 1). \quad (8.28)$$

This is the unramified calculation of [Gar15a, Theorem 2.1(3)]. Here $L_v(s, \pi_v, \mathrm{std} \otimes \chi_v)$ denotes the χ_v -twisted standard L -factor.

The character $\chi_v = \chi_{V,v}$ associated with the quadratic space $V/\mathbb{Q}$ is given by

$$\chi_v : \mathbb{Q}_v^\times \longrightarrow \{\pm 1\}, \quad \chi_v(a) = (a, (-1)^{m/2} \det(V))_v, \quad (8.29)$$

²We use the diagonal realization of the doubling construction, in which the Siegel parabolic stabilizes the diagonal Lagrangian; see [Gar15a, § 2.2].

where $m = \dim_{\mathbb{Q}} V$ is even and $(\cdot, \cdot)_v$ denotes the Hilbert symbol.³

Motivated by the unramified calculation, we introduce the following normalization; see [Gar15a, § 2.2, equation (2.12)].

Definition 8.7. For every place v of $\mathbb{Q}$, define the normalized local zeta integral by

$$Z_v^0(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v) := \frac{d_v(s, \chi_v)}{L_v(s + \frac{1}{2}, \pi_v \times \chi_v, \text{std})} Z_v(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v). \quad (8.30)$$

Here $d_v(s, \chi_v)$ is given by (8.28), also at $v = \infty$, using the usual archimedean local L -factors.⁴

By (8.27), normalized spherical data give

$$Z_v^0(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v) = 1. \quad (8.31)$$

In particular, this holds at all but finitely many places for the factorizable global data fixed above.

Write

$$\Lambda(s, \pi \times \chi_V, \text{std}) := \prod_v L_v(s, \pi_v \times \chi_v, \text{std})$$

for the completed twisted standard L -function, and set

$$d(s, \chi_V) := \prod_v d_v(s, \chi_v) = \Lambda\left(s + n + \frac{1}{2}, \chi_V\right) \prod_{j=1}^n \Lambda_{\mathbb{Q}}(2s + 2j - 1). \quad (8.32)$$

Here $\Lambda(s, \chi_V)$ is the completed Hecke L -function and

$$\Lambda_{\mathbb{Q}}(s) := \pi^{-s/2} \Gamma(s/2) \zeta(s)$$

is the completed Riemann zeta function; see [Gar15a, § 2.4, equations (2.36)–(2.38)].

The Euler factorization of the global doubling zeta integral can therefore be rewritten as follows.

Corollary 8.8

For factorizable data as above, one has the identity of meromorphic functions

$$Z(s, f_s, \varphi, \tilde{\varphi}) = \frac{\Lambda(s + \frac{1}{2}, \pi \times \chi_V, \text{std})}{d(s, \chi_V)} \prod_v Z_v^0(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v). \quad (8.33)$$

All but finitely many factors in the product of normalized local zeta integrals are equal to 1.

Proof. For $\text{Re}(s) \gg 0$, substitute (8.30) into the Euler factorization of Proposition 8.6; see also [GQT14, § 11.6]. The identity then extends by meromorphic continuation. $\square$

Remark 8.9

When $n = 2$, the local denominator is

$$d_v(s, \chi_v) = L_v\left(s + \frac{5}{2}, \chi_v\right) \zeta_v(2s + 1) \zeta_v(2s + 3), \quad (8.34)$$

and hence

$$d(s, \chi_V) = \Lambda\left(s + \frac{5}{2}, \chi_V\right) \Lambda_{\mathbb{Q}}(2s + 1) \Lambda_{\mathbb{Q}}(2s + 3). \quad (8.35)$$

Thus, for Sp_4 , formula (8.33) becomes

$$Z(s, f_s, \varphi, \tilde{\varphi}) = \frac{\Lambda(s + \frac{1}{2}, \pi, \text{std} \otimes \chi_V)}{\Lambda(s + \frac{5}{2}, \chi_V) \Lambda_{\mathbb{Q}}(2s + 1) \Lambda_{\mathbb{Q}}(2s + 3)} \prod_v Z_v^0(s, f_{s,v}, \varphi_v, \tilde{\varphi}_v). \quad (8.36)$$

³See [Gar15a, § 2.1, equation (2.1)]. In particular, if $\dim_{\mathbb{Q}} V = 4$, then $\chi_v(a) = (a, \det(V))_v$.

⁴At the real place, $\zeta_{\infty}(s) = \Gamma_{\mathbb{R}}(s) := \pi^{-s/2} \Gamma(s/2)$ and $L_{\infty}(s, \chi_{\infty}) = \Gamma_{\mathbb{R}}(s + \epsilon)$, where $\chi_{\infty} = \text{sgn}^{\epsilon}$ and $\epsilon \in \{0, 1\}$.

8.3 The Hodge–Deligne inner product formula

Theorem 8.10 (Hodge–Deligne inner product formula)

We have

$$\langle r_D(\Theta^{\text{mot}}(\omega_g)), \Theta^B(\omega_{g'}) \rangle = \frac{\prod_{v \in S} Z_v^0(-1/2, f_{-1/2, v}, g_v, g'_v)}{L(-1, \chi_V) \zeta(2)} \lim_{s \rightarrow 1} \frac{L(s, \pi \times \chi_V, \text{std})}{\zeta(s)}. \quad (8.37)$$

Here S is a finite set of places of $\mathbb{Q}$ containing the archimedean place and every finite place at which π_v , χ_v , or the chosen local data are ramified. Outside S , the local data are normalized spherical.

Proof. This follows from Propositions 8.2, 8.3, and 8.6. $\square$

Remark 8.11

- The quotient in (8.37) is understood meromorphically at $s = 1$. The pole of $\zeta(s)$ is cancelled by the corresponding factor in $L(s, \pi \times \chi_V, \text{std})$, so the displayed limit is finite.
- This formula is an analogue of the Rallis inner product formula and its geometric and arithmetic counterparts.

Theorem 8.12 (Beilinson-type regulator formula)

We have the following Beilinson-type regulator formula:

$$\langle r_D(\Theta^{\text{mot}}(\omega_g)), \Theta^B(\omega_{g'}) \rangle = \frac{\prod_{v \in S} Z_v^0(-1/2, f_{-1/2, v}, g_v, g'_v)}{L(-1, \chi_V) \zeta(2)} L(1, \Pi, \text{std}). \quad (8.38)$$

Proof. For a finite place v , let ϕ_{π_v} and ϕ_{Π_v} denote the L -parameters of π_v and Π_v , respectively. By [AG17, Theorem D.2(i)(b), (ii)(c)], they satisfy

$$\phi_{\pi_v} \otimes \chi_{V, v} = \phi_{\Pi_v} \oplus \mathbf{1}, \quad (8.39)$$

and consequently

$$L(s, \pi \times \chi_V, \text{std}) = \zeta(s) L(s, \Pi, \text{std}). \quad (8.40)$$

The asserted regulator formula now follows from Theorem 8.10. $\square$

Remark 8.13

- For certain types of V , Beilinson-type regulator formulas, also called complex regulator formulas, have been obtained using modular units. The following table summarizes the results according to the classification of V .

Let $V/\mathbb{Q}$ have dimension 4 and real signature $(2, 2)$. Write $r = \text{witt}_{\mathbb{Q}}(V)$ and $d = \text{disc}(V)$. The table lists complex regulator formulas with trivial coefficients; groups are given up to isogeny.

r	d	$\mathrm{SO}(V)^a$	Regulator literature
2	1	$\mathrm{SL}_2 \times \mathrm{SL}_2$	Rankin–Selberg: Beilinson [Bei85a]; BDR [BDR15]; ^b LLZ [LLZ14].
0	1	$\mathrm{SL}_1(D) \times \mathrm{SL}_1(D)$	Rankin–Selberg: Ramakrishnan [Ram86]. ^c
1	$\neq 1$	$\mathrm{Res}_{F/\mathbb{Q}} \mathrm{SL}_2$	Asai: LLZ [LLZ18] (open surface). ^d
0	$\neq 1$	$\mathrm{Res}_{F/\mathbb{Q}} \mathrm{SL}_1(B)$	Asai: none known.

^aFor $d \neq 1$, $F = \mathbb{Q}(\sqrt{d})$. The quaternion division algebras $D/\mathbb{Q}$ and B/F are split at every real place.

^bBDR assumes $\chi_f \chi_g \neq 1$.

^cNot an explicit cycle construction; see [Gar15b].

^dFor the current-pairing gap and its repair, see [BGCLRJ24b]. LLZ’s compactification argument does not cover weight $(2, 2)$.

- Our construction also covers compact orthogonal Shimura varieties, beyond the reach of the modular-unit approach.
- The method treats all four cases uniformly.
- Our method can be generalized to higher-dimensional Shimura varieties, whereas the modular-unit approach is not suited to that setting.

8.4 Nonvanishing of the motivic theta lift

Theorem 8.14

For suitable choices of the finite Schwartz function φ_f and the finite component g_f of $g = g_f \otimes g_\infty$, the motivic theta lift $\Theta^{\mathrm{mot}}(\omega_g)$ is nonzero. Thus the construction gives a nonzero motivic class in $\mathrm{CH}^2(\mathrm{Sh}_{\mathrm{SO}(V)}, 1)$.

Proof. It follows from Theorem 8.12 that

$$\langle r_D(\Theta^{\mathrm{mot}}(\omega_g)), \Theta^B(\omega_{g'}) \rangle = \prod_{v \in S} Z_v^0(-1/2, f_{-1/2,v}, g_v, g'_v) \frac{L(1, \Pi, \mathrm{std})}{L(-1, \chi_V) \zeta(2)}. \quad (8.41)$$

First, $L(-1, \chi_V) \neq 0$ and $\zeta(2) \neq 0$. By [GQT14, Theorem 1.4(i)(b)], the nonvanishing

$$\theta_{\mathrm{Sp}_4}(\Pi) \neq 0, \quad \theta_{\mathrm{SL}_2}(\Pi) = 0 \quad (8.42)$$

implies

$$L(1, \Pi, \mathrm{std}) \neq 0. \quad (8.43)$$

Next, at each finite place v , the nonvanishing of the local doubling zeta integral allows us to choose $\varphi_f, \varphi'_f, g_v$, and g'_v such that

$$Z_v^0(-1/2, f_{-1/2,v}, g_v, g'_v) \neq 0. \quad (8.44)$$

Finally, for the chosen archimedean test vectors $f_{-1/2,\infty}, g_\infty$, and g'_∞ , [Gar15b, p. 35] gives

$$Z_\infty\left(-\frac{1}{2}, f_{-1/2,\infty}, g_\infty, g'_\infty\right) \neq 0. \quad (8.45)$$

□

Remark 8.15

- The nonvanishing of these motivic classes suggests a possible route to Euler systems, which we hope to pursue in future work.

- For $(\mathrm{Sp}_{2n}, \mathrm{SO}(2n - 2, 2))$, the proposed route is to combine the global argument with the archimedean local theta correspondence.

Future work

- (1) When n is odd, the half-integral power $\omega^{1+n/2}$ must be interpreted on the metaplectic stack; see [Can16].
- (2) The construction of $\Theta^{\text{mot},1}$ extends to the general reductive dual pair $(\text{Sp}_{2g}, \text{SO}(V))$ when $g > 3$, where V has signature $(n, 2)$, provided that the following cohomological vanishing holds:

$$H^{\frac{g(g+1)}{2}-3}(X^{\text{tor}}, \omega^{g-n/2}) = 0. \quad (9.1)$$

The available vanishing theorem, however, only gives the range

$$i > \frac{g(g+1)}{2} - 2, \quad (9.2)$$

which does not imply (9.1).

- (3) To integrate Borcherds products into the general construction of $\Theta^{\text{mot},1}$, we are reformulating their construction in adelic and geometric terms [QSZ].
- (4) When $\dim V = 2g - 2$, relating the Beilinson regulator of the motivic theta lift to non-critical L -values requires generalizing the constructions in Sections 6.5.2 and 7.5 to arbitrary g .
- (5) Proving nonvanishing of the motivic theta lift for general g also requires a nonvanishing result for the chosen archimedean test vectors.

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